



Diffusion Models in Robotics: A Survey

Xiaokang Liu¹ · Kevin Yuchen Ma¹ · Chen Gao¹ · Mike Zheng Shou^{1,2}

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Abstract

Diffusion models have demonstrated outstanding performance in image and video generation over recent years. Their strong expressive power and stable training dynamics have recently motivated researchers to integrate them into robot learning. In this survey, we trace and synthesize the latest progress in diffusion-based robot learning. We begin with an in-depth overview of robotics research history and the fundamentals of diffusion models. Next, we systematically examine diffusion-based methods in robotics, including diffusion policy approaches and diffusion-based image and video synthesizer. For diffusion policy-based methods, we present a comprehensive, top-down analysis of related approaches. We then explore the application of diffusion-based image and video generation techniques in robotic tasks, covering areas such as planning and data collection. Following this, we provide a review of key benchmarks and datasets, then summarize the major objectives addressed by diffusion-based methods and analyze their common strategies. Finally, we discuss the current challenges and outline promising future directions for diffusion-based robotic learning. Through this survey, we aim to provide a thorough resource inspiring further research at the intersection of diffusion modeling and robotics.

Keywords Diffusion models · Computer vision · Robotics · Robot learning

1 Introduction

In the field of generative artificial intelligence, diffusion models (Ho et al., 2020) have rapidly emerged in the past few years as a powerful solution. Based on an iterative denoising process, these models learn to progressively reverse a noising procedure to transform a simple random distribution into complex, high-fidelity samples. Their strong probabilistic modeling ability and remarkable sample quality have fos-

tered success in various generation domains, including image synthesis (Lugmayr et al., 2022), video generation (Ho et al., 2022) and audio generation (Kong et al., 2020), demonstrating superior performance compared to previous generative models such as Variational Autoencoders (VAE) (Kingma et al., 2013) and Generative Adversarial Networks (GAN) (Goodfellow et al., 2014).

Robotics is a dynamic field devoted to designing agent systems that perceive, plan, and interact in the physical world. Over recent decades, robot learning has emerged as a central approach, with foundational works in reinforcement learning (RL) and imitation learning (IL) enabling robots to acquire complex behavior through trial-and-error and demonstration (Hua et al., 2021). However, due to the lack of vast amounts of high-quality real-world robotic data, previous methods often struggle to capture the inherently multi-modal nature of robotic tasks, and frequently yield imprecise or unstable control in complex scenarios. Such limitations often hinder the deployment of generalizable and robust robotic systems, marking the need for novel techniques.

Building on the success of diffusion models in generative tasks, researchers have recently begun integrating these approaches into robotics to tackle long-standing challenges in robot learning. As shown in Fig. 1, a key innovation is the

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✉ Mike Zheng Shou
mike.zheng.shou@gmail.com

Xiaokang Liu
xliu@u.nus.edu

Kevin Yuchen Ma
kevinskwk@gmail.com

Chen Gao
gaochen.ai@gmail.com

¹ Department of Computer Science, National University of Singapore, 13, Computing Dr, Singapore 117417, Singapore

² Department of Electrical and Computer Engineering, National University of Singapore, 4 Engineering Drive 3, Singapore 117583, Singapore

development of diffusion policies (Chi et al., 2023), which leverage the probabilistic framework of diffusion models to generate robust, multimodal representations of action-space data—such as robot action trajectories or end-effector poses. This approach enables a more faithful capture of the high-dimensional, multimodal distributions inherent in robotic tasks, ultimately enhancing model stability and adaptability. Concurrently, diffusion models originally designed for pixel-space data synthesizing such as images and videos, have been repurposed within robotics to synthesize training data or task goals (Du et al., 2023b; Yu et al., 2023). These methods seek to provide a solution to data scarcity challenges in robotic learning, further underscoring the transformative potential of diffusion-based techniques.

While the intersection of diffusion modeling and robotics has seen rapid growth, there have been some specialized reviews such as Wolf et al. (2025), which provides a focused analysis of diffusion techniques within robotic manipulation. However, the field lacks a holistic framework that encompasses the full diversity of robotic embodiments and applications. This survey distinguishes itself by moving beyond the confines of static manipulation, expanding the discourse to include dynamic domains such as robot navigation, mobile manipulation, and the handling of deformable objects. Furthermore, rather than adopting a purely application-driven taxonomy, we propose a three-tier hierarchical taxonomy that analyzes current progress at the algorithmic, architectural, and modular levels for diffusion policies. This structured perspective allows researchers to identify the integration between generative modeling and robotic control that transcend specific task boundaries, providing a unified roadmap for future generalist robotic agents.

In this survey, we focus on the role of diffusion-based models in robotics, breaking down their applications across several major areas. We examine the core components of diffusion policies for action-space data generation and pixel-space diffusion generators for planning and data augmentation. In addition, we review the various objectives that current diffusion-based methods aim to address. Through this survey, we hope to provide a comprehensive overview of the field, helping future researchers—whether from a diffusion modeling or robotics background—gain insights and inspiration from recent advancements.

This survey is organized as follows. We begin by introducing the research background in robotics and the fundamentals of diffusion models (Sect. 2). Next, we provide a comprehensive review of diffusion policy-based approaches, breaking down their components from a hierarchical perspective and analyzing the main learning objectives of current research (Sect. 3). Following this, we examine the use of image and video diffusion models in robotics (Sect. 4). We then present the mainstream benchmarks and datasets commonly used in the robotics field (Sect. 5). We also review several

major objectives and target tasks addressed by diffusion-based methods in robotics (Sect. 6). Finally, we discuss key challenges and outline potential future research directions for diffusion-based robotic learning (Sect. 7).

2 Preliminaries

2.1 Evolving of Robotics Approaches

Traditional robotic systems were structured around a hierarchy of modules, including perception, planning, and control, each primarily driven by mathematical models and rule-based strategies. These methods provided reliable and interpretable solutions for structured environments, but they struggled with uncertainty, adaptation, and high-dimensional state spaces (Alterovitz et al., 2016). Foundational algorithms such as Rapidly Exploring Random Trees (RRT) (Lavalle, 1998) and Linear Quadratic Regulator (LQR) (Lehtomaki et al., 1981) have been widely used in robot planning and control. However, their reliance on precise modeling and extensive hand-engineered rules made them less effective in unstructured or dynamic settings.

Before the rise of deep learning, data-driven methods had been introduced in robotics to acquire skills from experience. Early supervised learning has been used in perception tasks, such as object recognition and localization. Behavior cloning, or imitation learning, trained policies by mapping states to actions, but suffered from compounding errors during execution (Pomerleau, 1988). Reinforcement learning emerged as a promising paradigm, where policies were learned through trial and error. However, early RL methods struggled with sample inefficiency and high-dimensional control problems, limiting their practical applications (Kaelbling et al., 1996).

Recent progress in deep learning has improved perception, control, and decision-making in robotics. Vision models pre-trained on large-scale datasets, *e.g.*, ResNet (He et al., 2016), CLIP (Radford et al., 2021), and R3M (Nair et al., 2022), have been widely used to extract visual representations for downstream robotics tasks. Deep Reinforcement Learning has enabled robots to learn complex tasks in high-dimensional spaces, exemplified by Kalashnikov et al. (2018). Behavior cloning has advanced with large-scale datasets, mitigating some of its limitations by techniques like dataset aggregation (Ross et al., 2011). Furthermore, Vision-Language-Action (VLA) models (Brohan et al., 2023; O’neill et al., 2023; Kim et al., 2024a) have introduced multimodal learning capabilities, enabling robots to leverage textual and visual inputs for task execution. Despite these advancements, most learning-based methods struggle to handle multimodal distributions, and some face challenges such as training difficulty and lack of precision. Recently, the rise of diffusion policy (Chi et al.,

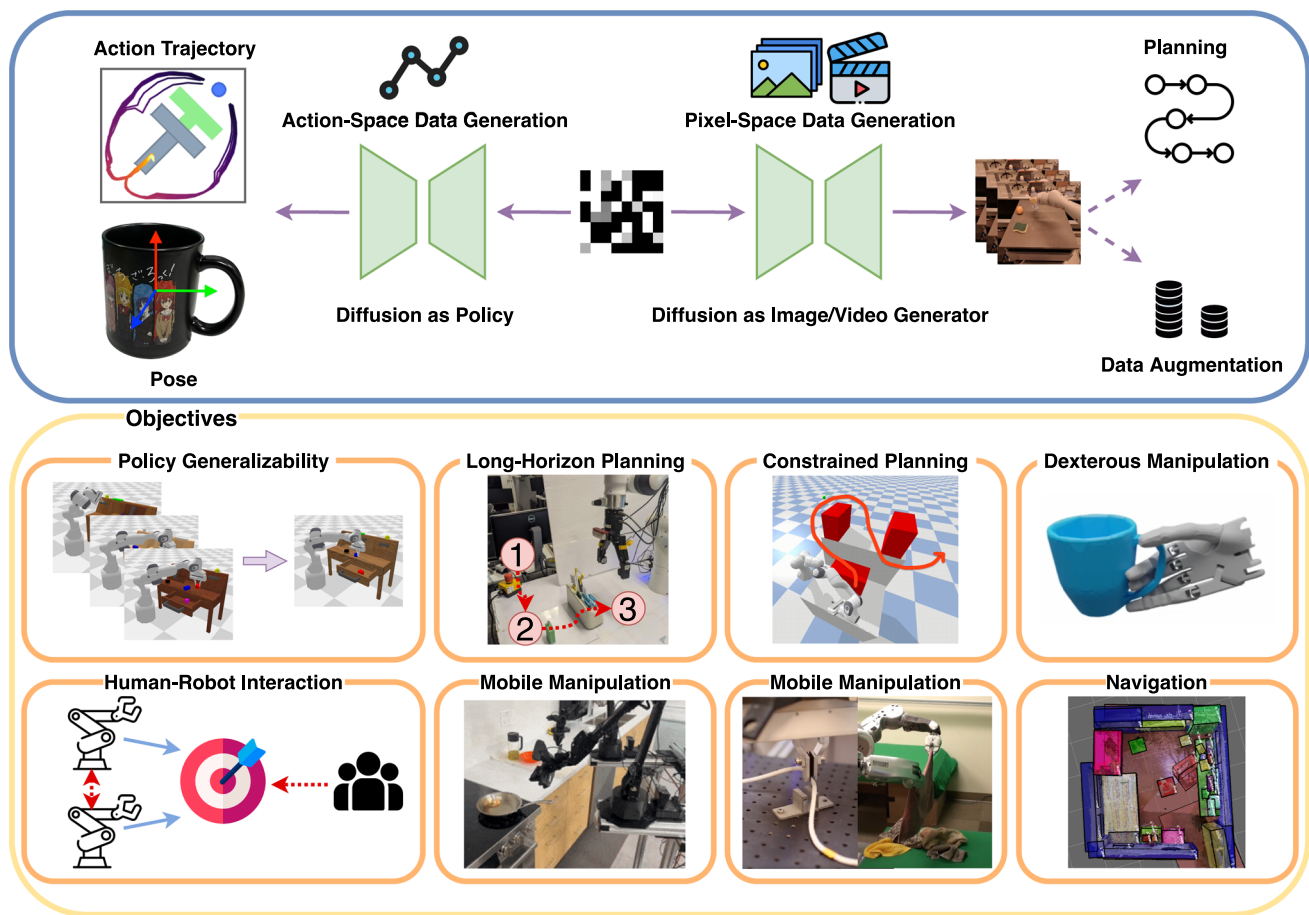


Fig. 1 An overview of diffusion models in robotics. The upper part illustrates diffusion models applied to both pixel-space and action space data generation. The lower part highlights the major objectives addressed by diffusion-based methods

2023) aims to address these issues by incorporating diffusion models into visuomotor policy learning.

2.2 Diffusion Models

Before discussing the application of diffusion models in robotics, it is crucial to understand the basic concepts of the diffusion process. Specifically, diffusion models demonstrate remarkable success in image and video generation tasks (Zhang et al., 2023; Zheng et al., 2024; Rombach et al., 2021), with prominent formulations including denoising diffusion probabilistic models (DDPM) (Ho et al., 2020), score-based generative models, and stochastic differential equations (Score SDEs) (Song et al., 2020). In the following, we leverage DDPM to elucidate the core of the diffusion process.

Concretely, DDPM represents a class of generative models that learn to generate data through an iterative denoising process. This kind of model operates by gradually corrupting training data with Gaussian noise in a forward process, then learning to reverse this corruption in a backward process

to generate new samples. Technically, the forward process is defined as a fixed Markov chain that gradually adds Gaussian noise to the data over T timesteps according to a predefined variance schedule $\{\beta_t\}_{t=1}^T$. Given a data point $\mathbf{x}_0$, the noise-corrupted sample $\mathbf{x}_t$ at timestep t is distributed as:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}). \quad (1)$$

This process allows for efficient closed-form sampling at any timestep t using the reparameterization trick:

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_0 + \sqrt{1 - \alpha_t}\epsilon, \quad (2)$$

where $\alpha_t = \prod_{i=1}^t (1 - \beta_i)$ and $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The generation process learns to reverse the diffusion through a learned Markov chain parameterized by θ . Starting from Gaussian noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the model iteratively denoises the sample as follows:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)), \quad (3)$$

where μ_θ and Σ_θ are learned neural networks that predict the mean and covariance of the reverse distribution. DDPMs are trained to minimize the variational lower bound on the negative log-likelihood. A key simplification (Ho et al., 2020) involves training a noise prediction network ϵ_θ to predict the noise ϵ added during the forward process:

$$\mathcal{L} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left[\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2 \right], \quad (4)$$

where t is uniformly sampled from $\{1, \dots, T\}$. With trained model ϵ_θ , we can generate new data $\mathbf{x}_0$ by first sampling from a Gaussian noise $\mathbf{x}_T$, then iteratively denoising it through $\mathbf{x}_{t-1} \sim p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ until t reaches 1.

The intrinsic nature of the diffusion process makes it suitable for robotics applications requiring sequential decision-making, as diffusion can capture complex multimodal distributions while maintaining stable training ability compared to other generative approaches.

3 Diffusion as Policy

In this section, we provide a comprehensive review of contemporary methods employing diffusion policy (Chi et al., 2023) in robotic tasks. We begin with an overview of diffusion policy, followed by a break down of diffusion policy methods, providing an in-depth analysis of various technical approaches. After this we discuss two learning objectives commonly tackled in diffusion policy methods. Figure 2 shows an outline.

3.1 Overview of Diffusion Policy

Inspired by advancements in diffusion models within the fields of image and video generation, Chi et al. (2023) introduce the concept of diffusion policy by applying the diffusion process to action trajectory generation, marking the first application of this approach in robotics.

Policy learning from demonstration is often formulated as a regression problem, mapping environment observations to corresponding actions. However, traditional methods may struggle with the multimodal nature and precise action control required in robotics applications. Diffusion models, with their strong generative capabilities and powerful distributional expressivity, offer a promising alternative for robot policy learning.

As illustrated in Fig. 3, a general framework of diffusion policy is designed to take state observation O_t at timestep t as input, and generate clips of action trajectory $\mathbf{a}_t = (a_t, a_{t+1}, \dots, a_{t+H})$. H denotes the prediction horizon. The generation process follows the same iterative

denoising procedure as that described in Sect. 2.2 with two key modifications:

- The denoising target, originally denoted as $\mathbf{x}$ in conventional diffusion models, is replaced by the action trajectory $\mathbf{a}_t$.
- The denoising process is conditioned on the observation O_t , enabling the policy to generate actions accordingly to the current state.

In summary, the diffusion policy conveys several advantages:

- **Modeling Multimodal Distributions.** By learning the gradient of the action distribution's score function, the diffusion policy naturally accommodates multiple valid action modes. The stochastic initialization and iterative refinement allow the policy to explore different convergence basins and model the multimodal behavior.
- **Precise Action Control.** Unlike traditional policy learning methods and discrete-token VLAs (Brohan et al., 2022) that quantize actions into finite bins, diffusion policies utilize a continuous action space. The iterative denoising process progressively shifts estimates toward high-probability action distributions to mitigate quantization errors. Although modern VLAs are closing this gap, this continuous representation remains a primary driver for fine-grained precision.
- **High-Dimensional Output Capability.** Diffusion models have demonstrated excellent scalability in generating high-dimensional outputs. In the context of robotics, this means that the policy can jointly predict a sequence of actions that maintains temporal consistency during execution.
- **Stable Training.** Diffusion models provide more stable training dynamics compared to other generative paradigms such as GANs or energy-based models. This stability stems from the use of a simple mean squared error loss on a noise-prediction objective, which provides a deterministic regression target throughout training (Ho et al., 2020). This avoids the challenges of non-convergence and mode collapse often associated with adversarial training (Gui et al., 2021; Du and Mordatch, 2019).

Different approaches vary in their configurations. The modality of the state observation O_t can differ, *e.g.*, RGB images and 3D point clouds, while the generated trajectory may be represented either as a set of keypoints or as a continuous sequence. Another important design choice involves selecting the historical and prediction horizons, which determine the number of frames considered. Furthermore, the architectures of the diffusion models can differ significantly,

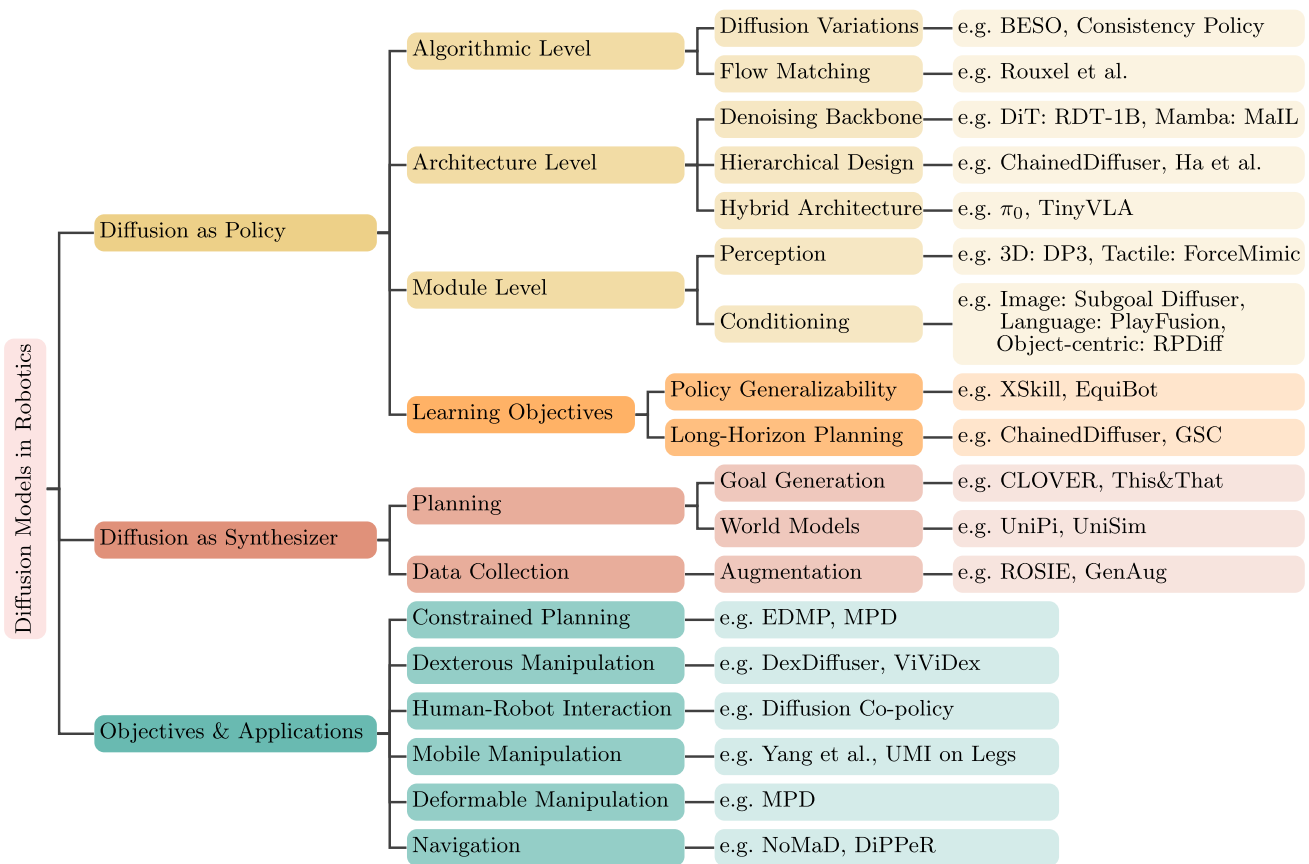


Fig. 2 The main content review of this survey about diffusion models in robotics

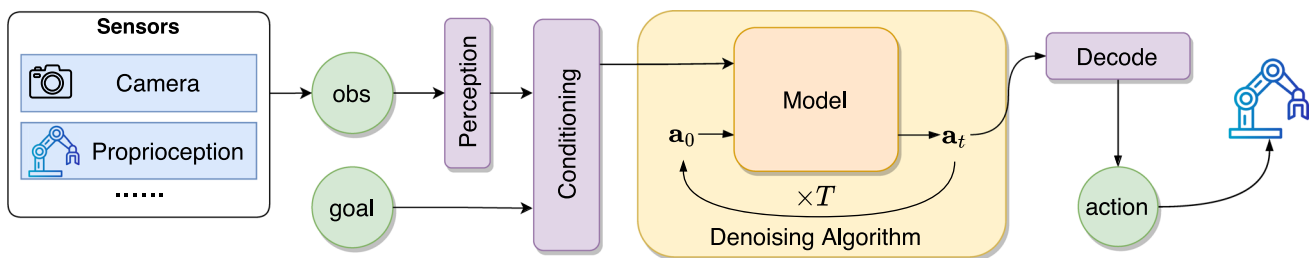


Fig. 3 An overall structure of diffusion policy methods in robotics

spanning from basic CNNs to more advanced Diffusion Transformer (DiT) (Peebles and Xie, 2023) models.

3.2 Technical Approaches

In this section, we break down the formation of diffusion policies to provide a hierarchical view of different methods' focuses. We analyze methods from three perspectives: the algorithmic level, which focuses on various diffusion process variations; the architecture level, which covers network and concrete model designs; and the module level, which examines the different perception and conditioning modules.

3.2.1 Algorithmic Level

At the algorithmic level, diffusion policies are founded on a range of mathematical frameworks that enable robust and multimodal action generation. This section mainly discusses how diffusion processes are harnessed in different approaches—whether through probabilistic modeling, flow matching or consistency enforcement—to form the backbone of robotic control strategies. While the most common practice is to use DDPM or DDIM as introduced in Sect. 2.2, recent advances in diffusion algorithms have given rise to many alternative approaches.

Diffusion Model Variations. One of the alternatives to DDPM is Score-Based Generative Models (SGMs) (Song et al., 2020), which differ by estimating a score in the continuous-time framework rather than using discrete denoising steps like DDPM/DDIM. This approach enables SGMs to employ more flexible sampling strategies, often resulting in higher sample quality and reduced diffusion steps, which enhances both computational efficiency and robustness in multimodal action generation. Representative methods employing SGMs in robotic diffusion policy include BESO (Reuss et al., 2023), MPD (Scheikl et al., 2024), and Guo et al. (Guo et al., 2024). In the field of image generation, some studies (Song et al., 2023) have proposed enforcing consistency between denoising steps through distillation, allowing consistency models to achieve comparable results with just one or a few denoising iterations. In the robotics domain, Consistency Policy (Prasad et al., 2024) and ManiCM (Lu et al., 2024) adapt these models, demonstrating significantly faster inference speeds while maintaining performance.

Flow Matching. Flow matching (Liu et al., 2022) is a recent generative technique that establishes a straight path between noise and data, yielding deterministic outputs. In contrast to traditional diffusion processes, flow matching has been shown to be numerically more stable and to accelerate the generation process. Several methods (Rouxel et al., 2024; Chisari et al., 2024; Zhang and Gienger, 2024; Zhang et al., 2024a; Funk et al., 2024) leverage flow matching as a robot action policy based on the intuition that its deterministic nature can benefit robotics models and also enables faster inference without performance degradation. Other methods extend the scope further—SRFMP (Ding et al., 2024) generalizes the flow matching policy to Riemannian manifolds, while FlowNav (Gode et al., 2024) applies it to navigation scenarios.

3.2.2 Architecture Level

At the architecture level, diffusion policies rely on diverse neural network structures to balance efficiency and adaptability. Common denoising backbones include CNNs, transformers, U-Net, and DiT, with newer models like Mamba improving scalability. Architecture design varies from coarse-to-fine strategies to skill-based representations, while constraint-aware conditioning ensures stable, guiding control. The following sections explore these components.

Denoising Backbone Compared to diffusion models in image and video generation, robotic action generation operates within a smaller latent space. As a result, many early-stage diffusion-based policies employ relatively simple CNN (Chi et al., 2023) or Transformer (Reuss et al., 2023) architectures as denoising backbones. While U-Net backbones have been used in some robotic diffusion policies

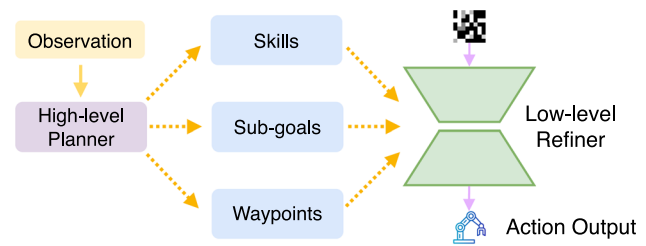


Fig. 4 Illustration of hierarchical design in diffusion policy

(Hou et al., 2024b) for their strong local temporal inductive biases and effective short-to-medium range generation, their hierarchical downsampling can introduce an inherent over-smoothing effect. Theoretically, this convolutional bottleneck may obscure the high-frequency control signals and sharp action transitions essential for millimetric precision (Chi et al., 2023). Consequently, contemporary research has moved toward using DiT (Peebles and Xie, 2023) and selective state-space models (Gu and Dao, 2023). These architectures avoid fixed convolutional receptive fields, enabling superior modeling of global dependencies and more faithful representation of high-rate action changes.

In addition to these fundamental architectural designs, various models have been proposed to address specialized problems. Given the spatial characteristics of robotic tasks, some methods (Wang et al., 2024c; Yang et al., 2024a; Ryu et al., 2024) propose to enforce equivariance in transformation domains such as rotation and scaling, enhancing policy generalization to unseen environments. Meanwhile, some other work (Jia et al., 2024; Cao et al., 2024) leverage Mamba (Gu and Dao, 2023) models in robotic manipulation, utilizing their capabilities of maintaining sequential modeling abilities with low computation complexity.

Hierarchical Design Diffusion policy approaches in robotics encompass a wide range of methodologies for designing effective architectures for robot control. A common focus is hierarchical trajectory generation, where complex tasks are addressed through layered planning and refinement, as illustrated in Fig. 4. In this section, we review several key hierarchical design frameworks that have been systematically adopted by many researchers.

One prevalent strategy is coarse-to-fine trajectory generation. In this approach, a high-level planner first schedules task-relevant key waypoints or sub-goals, which are then used to guide a low-level diffusion policy responsible for generating precise action trajectories. Several methods follow this design paradigm (Guo et al., 2024; Huang et al., 2024f; Ma et al., 2024), with ChainedDiffuser (Xian et al., 2023) serving as a representative example. Beyond waypoints generation, some approaches (Ajay et al., 2023; Hatch et al., 2024) extend the hierarchical framework by producing intermediate image sub-goals, which further guide low-level planning

and execution. These methods achieve better performance on long-horizon tasks by breaking them into fine-grained segments.

As a subset of hierarchical framework, many methods explore skill-based learning in diffusion policy. A common approach is to decompose the long-horizon tasks into a set of primitive skills, which can be either predefined (e.g., pick, push, place) or learned. PlayFusion (Chen et al., 2023) draws inspiration from VQ-VAE (Van Den Oord and Vinyals, 2017) and proposes learning a discrete skill codebook supplemented by language. It introduces a vector quantization bottleneck into the diffusion process, generating discrete skills that are subsequently decoded into action sequences. Similarly, XSkill (Xu et al., 2023) learns manipulation skills prototypes by aligning human and robot demonstration videos. During inference, it extracts a skill sequence from human video and maps it to robot actions. Many approaches also leverage language models as high-level planners to generate textual skill instructions. Ha et al. (2023) and SkillDiffuser (Liang et al., 2024c) both prompt a language model to decompose instructions into a sequence of primitive task descriptions, then learn a skill codebook accordingly for low-level control. Additionally, GSC (Mishra et al., 2023) trains an independent diffusion model for each primitive skill and chains them sequentially using probabilistic methods.

Hybrid Architectures

Recently, in pursuit of both high-level planning and low-level precision, some methods have explored hybrid architectures that integrate diffusion process with VLA models. In these hybrid architectures, the VLA backbone—often a high-parameter transformer pretrained on internet-scale data—processes multimodal inputs to generate high-level semantic action tokens. These tokens are then fed into a diffusion model head through conditioning mechanisms, where the diffusion component is responsible for refining these abstract representations into precise, continuous action trajectories for robot execution (Kim et al., 2025; Black et al., 2024; Team et al., 2024).

More concretely, this hybrid paradigm addresses the frequency gap between large-scale reasoning and real-time control. While the VLA backbone provides the knowledge for complex task understanding, its massive parameter count often leads to big inference latency. Diffusion heads serve as high-frequency visuomotor controllers that interpret the VLA's outputs into fine-grained signals (Black et al., 2024; Wen et al., 2024). In addition to faster inference, another primary advantage of this approach is that while traditional VLA models often rely on discrete action tokenization, the diffusion process operates in a continuous space, thus providing more precise control. This combination ensures that the system maintains the generalization ability of a foundation

model while achieving the precision required for contact-rich manipulation.

3.2.3 Module Level

At the module level, we focus on the design of critical components outside the primary denoising network, such as perception modules and additional conditioning modules. These modules process sensory data or integrate task-specific information to better steer the diffusion process, ultimately enhancing the performance of control policy.

Perception. In robotic tasks, systems perceive their environment using observational data from various sensors, including RGB and depth cameras, LiDARs, and tactile sensors. Different methods have explored diverse approaches to process and represent observational data. The encoded observation features are usually used as condition for diffusion policy guiding.

(a) RGB-D images perception. The most basic perception modules rely solely on RGB images to capture environmental information, some further leveraging depth information. Depending on the approach, one may utilize multiple cameras to gain a more comprehensive view of the environment (Hu et al., 2024; Ma et al., 2024) or incorporate several historical frames to ensure more stable control.

(b) 3D perception. In order to obtain more compact perception information, 3D representations are introduced to better reflect the real-world environment. These methods typically take RGB-D images as input and generate point clouds through pre-processing. For example, 3D Diffusion Policy (Ze et al., 2024), RISE (Wang et al., 2024b) and some other method (Liu et al., 2024b) use single-view point clouds, leveraging MLP-based or transformer-based encoder to compute 3D features. 3D Diffuser Actor (Ke et al., 2024) supports integrating data from multi-view cameras into a 3D scene token set. Some methods also explore integrating semantic information into point clouds. For instance, GenDP (Wang et al., 2024e) maps 3D semantic fields onto object point clouds, aiding to build correspondences with reference objects. Another line of work focus on object-centric manipulation. These methods (Li et al., 2024; Huang et al., 2024a) usually crop or extract the object point clouds from the observations and condition the policy based on the object's shape, geometry and orientation. In addition, some navigation models, such as DTG (Liang et al., 2024b), adopt LiDAR to capture comprehensive 3D surrounding information.

(c) Tactile perception. Tactile sensors provide force and positional information for robotic systems, enabling more subtle and accurate manipulation. Due to the diversity of current tactile sensors, perception techniques also vary. Two major types of tactile sensors are image-based and force-based. The image-based tactile sensors include GelSight (Yuan et al., 2017), which outputs RGB strain map images in

which brightness indicates depth and colors indicate tangential strains. Force-based sensors typically employ embedded pressure sensors and strain gauges to directly measure contact forces, with some converting them into discrete patches.

Various methods have been proposed to utilize tactile information to aid diffusion policy learning. VITaL Pretraining (George et al., 2024) aligns the visual and tactile latent spaces through contrastive learning, thereby enhancing the model's ability to comprehend both modalities. ForceMimic (Liu et al., 2024b) leverages hybrid imitation learning by training the diffusion policy to predict both action data and the corresponding force data, which enables robust performance in contact-rich manipulation. 3D-ViTac (Huang et al., 2024a) introduces a novel visuo-tactile learning framework. Instead of separately inputting visual and tactile data into the policy, it fuses data from the two modalities on the point cloud for policy conditioning. RDP (Xue et al., 2025) employs a hierarchical architecture, using a diffusion policy to predict high-level low-frequency action chunks and a low-level tactile feedback control system. Many other methods explore leveraging tactile data through MLP projection (Wu et al., 2024c), FFT encoding (Hou et al., 2024a), canonical representation (Wu et al., 2024a), GRU decoder (Xue et al., 2025) and self attention aggregation with other modalities (Aburub et al., 2024). Furthermore, Lin et al. (Lin et al., 2024) leverage fingertip tactile information from dexterous hands and combine it with other observation data to condition the diffusion policy. In summary, current tactile sensors are diverse with inconsistent standards, and most existing methods largely rely on their hardware design.

In conclusion, integrating various perception modalities into control policy learning is a growing trend. With advances in 3D vision, it is increasingly common to combine 3D visual perception with tactile input as complementary information. Future research can focus on improving the aggregation and collaboration of different perception modules.

Conditioning. A central design consideration in diffusion policy methods is the conditioning strategy, which directly influences how well the learned policy integrates both observational data and task-specific goals. Vanilla diffusion policy (Chi et al., 2023) conditions the model solely on observation images using techniques like FiLM (Perez et al., 2018); recent approaches start to explore diverse conditioning sources, incorporating both current observations and desired goals in various forms, as indicated in Fig. 5.

(a) Image goal conditioning. Goal-conditioning is a widely used technique in imitation learning, where each demonstration is augmented with goal states that indicate the target object or desired outcome after execution. The most common form of goal states is an image, which can represent the robot's final state, the target object, or other things that may help policy achieve the goals. BESO (Reuss et al., 2023) is the first to integrate goal-conditioning into diffusion policy,

using classifier-free guidance to simultaneously learn both goal-dependent and goal-independent policy. Furthermore, except for final goal conditioning, in order to maintain the system stability in long-horizon planning, some work decomposes the long tasks into a sequence of sub-goals (Huang et al., 2024f, d; Hatch et al., 2024).

However, a key challenge is that during real-world inference, the goal-state image after execution is not directly available. To address this, a line of research (Hu et al., 2023; Ko et al., 2023; Bu et al., 2024) explores leveraging image generation models to synthesize goal images for policy conditioning. We will further discuss them in Sect. 4.

Another direction is to enable robotic systems to directly learn from existing human demonstrations to complete specific tasks. Representative works include Track2Act (Bharadhwaj et al., 2024) and Dreamitate (Liang et al., 2024a), both of which propose tracking the key point movements of manipulated objects in videos and using this information to infer the robot's end-effector poses. Track2Act leverages a diffusion model to denoise the point trajectories from a randomized initial state, while Dreamitate learns to generate human demonstration videos with easily trackable tools. ViViDex (Chen et al., 2024b) extracts human hand and object poses in 3D trajectory representations, and retargets the hand poses to dexterous poses.

(b) Language goal conditioning. Compared to images, language provides a more interpretable and precise way to specify goal states, making it another intuitive tool for human-robot interaction. Recently, a wide range of studies have explored language-guided learning in robotics, with many focusing on text-driven generation ability of the diffusion models. PlayFusion (Chen et al., 2023) trains a skill-based diffusion policy in a learning-from-play setting, where each trajectory sequence is paired with a language instruction. The model encodes the language goal to condition the diffusion policy on the given text instruction. MDT (Reuss et al., 2024) leverages both image and language goal specifications, using a multimodal encoder to integrate them with observations for policy conditioning. ChainedDiffuser (Xian et al., 2023) tokenizes language instructions and 3D visual representations into a shared token space, which further enables both macro-action planning and local trajectory generation.

With recent advancements in VLMs, a line of research has explored their use as high-level planners. Ha et al. (2023) and DISCO (Hao et al., 2024) both leverage VLMs' reasoning capabilities to generate language-based plans of action primitives, while a diffusion policy conditioned on these action texts generates low-level trajectories. In addition, many works (Wen et al., 2024; Pan et al., 2024; Black et al., 2024; Xu et al., 2025) build directly on VLA models, first leveraging VLA to generate action tokens and then using

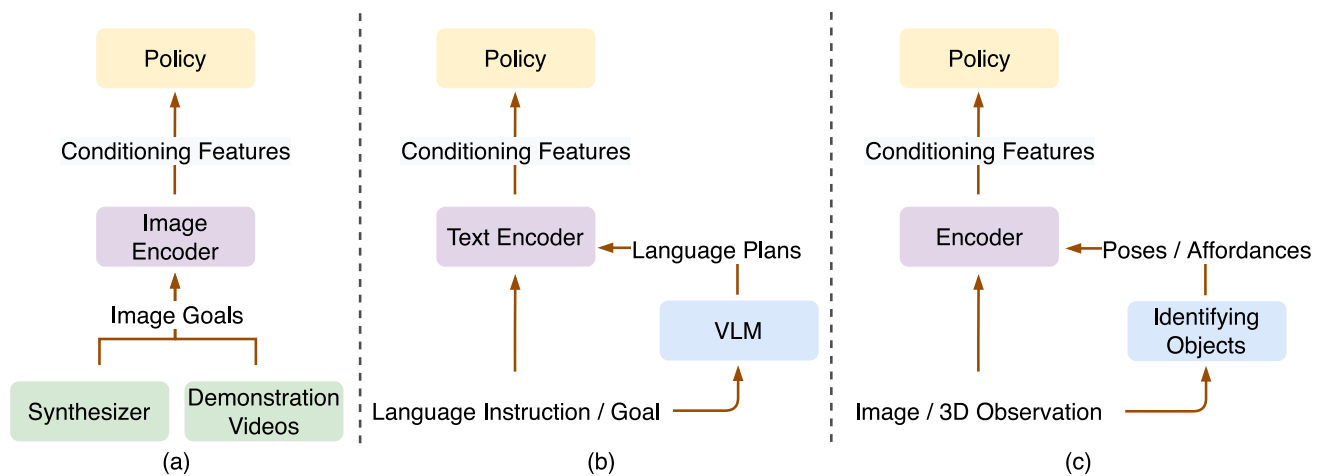


Fig. 5 Different condition strategies. (a) Image goal condition. (b) Language goal condition. (c) Object-centric condition

these tokens to condition a diffusion head for fine-grained trajectory generation.

(c) Object-centric conditioning. Object-centric manipulation with diffusion policy focuses on an object's shape, orientation, and interactions with its surroundings, modeling how these factors influence policy learning. One common approach in object-centric learning is tracking object poses and mapping them to the robot arm for execution. These methods (Urain et al., 2023; Hsu et al., 2024; Su et al., 2024; Chang et al., 2024) typically begin with pose estimation, leveraging diffusion models to denoise target object poses from an initial Gaussian noise. They generally use kinematic approaches to decode robot arm actions based on the object's initial and target poses. Some research (Liu et al., 2022; Simonov et al., 2023; Mishra, 2024) follows a similar approach to explore the reorientation task, where multiple objects must be rearranged into a specific configuration.

Another line of work focuses on affordance learning, which involves identifying functional parts of 3D objects (e.g., handles, straps) and using them to facilitate action prediction. CPM (Liu et al., 2024a), Lan-o3dp (Li et al., 2024; Rana et al., 2024) employ segmentation algorithms to extract object parts and train diffusion policies conditioned on object part poses to predict action trajectories. Additionally, some methods (Wang et al., 2024e; Ikeda et al., 2024; Chen et al., 2024) leverage the visual-semantic representation of 3D objects to help policies identify meaningful parts for manipulation. HDP (Wang et al., 2024d) introduces objective contacts, which indicates the interaction between objects and the gripper, to aid trajectory generation.

3.3 Reinforcement Learning with Diffusion Policy

While imitation learning currently dominates the landscape of diffusion-based robotics, pure behavioral cloning is fun-

damentally limited by compounding errors and a lack of robustness to out-of-distribution states. RL provides a pathway to overcome these constraints by optimizing policies against environmental rewards rather than static demonstrations. In the offline RL setting, diffusion models are particularly effective at the stitching problem—the ability to combine successful sub-segments from diverse, suboptimal data to solve complex goals. For instance, Decision Diffuser (Ajay et al., 2022) and related frameworks for stitching sub-trajectories (Kim et al., 2024b) leverage conditional diffusion to synthesize high-return paths by treating the planning task as a sequence-level denoising process. Furthermore, IDQL (Hansen-Estruch et al., 2023) demonstrates that utilizing a diffusion policy for offline RL improves OOD generalization by maintaining the expressive power of the action distribution while staying within the support of the behavior dataset.

To bridge the gap between offline pretraining and real-world execution, researchers have introduced methods for finetuning diffusion actors via online interaction. Diffusion Policy Policy Optimization (DPPO) (Ren et al., 2024a) adapts pretrained diffusion policies using policy gradient methods, allowing the robot to refine its movements based on sparse or dense reward feedback. This fine-tuning is critical for robotic manipulation tasks where the precision required for success often exceeds what can be learned solely from limited human demonstrations. By allowing the diffusion process to explore and optimize within the environment, agents can correct for the drift and distributional shifts that typically plague pure BC models.

Beyond serving as the policy backbone, diffusion models are increasingly utilized to synthesize other components of the RL loop, such as reward functions and world models. Diffuser (Janner et al., 2022) introduces a paradigm for flexible behavior synthesis, where a return-guided diffusion process enables long-horizon planning across diverse

robotic tasks. Concurrently, the challenge of reward engineering is being addressed by generative approaches like Diffusion Reward (Huang et al., 2024e), which learns reward functions directly from conditional video diffusion models. Despite these advancements, it is important to note that much of the current RL-based research for diffusion policies is still concentrated on simulation environments, such as maze navigation and trajectory generation. The application of these hybrid RL-diffusion techniques to high-dimensional, real-world robotic action generation remains an on-going area of study with significant potential for future growth.

3.4 Learning Objectives

In the pursuit of more robust and generalist robotic models, researchers have explored ways to enhance diffusion policies with more powerful capabilities. In this section, we discuss two major learning objectives in diffusion policy-based methods: policy generalizability and long-horizon planning. We examine how different approaches have been developed to address these challenges and advance the capabilities of diffusion policies.

3.4.1 Policy Generalizability

Generalizability remains a significant challenge in robotic manipulation, particularly when adapting to new tasks and environments with only a few demonstrations. Compared to VLAs (Brohan et al., 2022, 2023; Kim et al., 2024a; Black et al., 2024), which are trained on an extensive corpus of data, diffusion policy is often used in imitation learning with few demonstrations. To improve the generalizability of diffusion policy, various approaches have been proposed to address two questions: (1) *how to train a robust policy with fewer demonstrations?* (2) *how to adapt the policy to unseen tasks and environments?*

In the original diffusion policy (Chi et al., 2023), training a robust model for each task typically requires several hundred demonstrations. To reduce this requirement, some approaches leverage observation modalities with denser information. For instance, 3D Diffusion Policy (Ze et al., 2024) uses compact 3D representations encoded from point cloud rather than traditional 2D images to condition the diffusion model. Similarly, 3D Diffuser Actor (Ke et al., 2024) also leverages 3D visual data, but is conditioned on a tokenized 3D scene representation derived from multi-view camera inputs. In addition to plain 3D visual data, GenDP (Wang et al., 2024e) incorporates 3D semantic fields, embedding richer spatial information to help the policy focus on semantically meaningful parts of the object. This allows it to generalize across different instances within a category more effectively. Other methods include using linguistic guiding for 3D keyframes generations (Hao et al., 2024), and deploy-

ing a 3D spatially equivariant training strategy (Ren et al., 2024b), both showing the improvements of generalizability brought by 3D modality.

Drawing inspiration from molecular generation research (Xu et al., 2022; Wang et al., 2024c), Equivariant Diffusion Policy (Wang et al., 2024c) enforces equivariance in symmetric domains. By designing a diffusion model that is inherently rotationally equivariant—meaning any point cloud or robot state will yield the same output even if rotated—the model can learn to handle a wider range of cases with the same amount of demonstrations. Similarly, EquiBot (Yang et al., 2024a) and Diffusion-EDFs (Ryu et al., 2024), also based on equivariant diffusion model, employs a equivariant architecture, ensuring the learned policy is invariant to changes in rotation and translation, thereby improving data efficiency.

Other work such as CAGE policy (Xia et al., 2024) introduces causal attention in processing visual observation tokens. A set of learnable tokens cross-attends to the observation tokens, and the output tokens are then processed by causal transformers before being used as conditions for the diffusion model. JUICER (Ankile et al., 2024) as a data-efficient imitation learning pipeline, uses a small set of demonstrations to identify and annotate ‘bottleneck states’, synthetically generating data around these critical states. The augmented dataset helps to enable training a more robust policy with few original demonstrations.

3.4.2 Long-Horizon Planning

Despite recent progress in robotic imitation learning, the learned policy often fail beyond their training distribution due to unseen scenarios or self-induced compounding prediction errors during long-term execution (Ross and Bagnell, 2010). This limitation is particularly pronounced in manipulation tasks, where success rate depends on the robot’s ability to reason over extended time horizons, adapt to dynamic environments, and recover from unexpected perturbations.

A common approach for long-horizon planning is to split the overall action sequence into smaller segments and optimize each segment in a coarse-to-fine manner as described in Sect. 3.2.2. GSC (Mishra et al., 2023) proposes to decompose a long planning process into a sequence of primitive skills which can be modeled by the diffusion policy. The approach first trains separate skill diffusion models, then chains them together using predefined task skeletons when faced with a new task. Because each skill model focuses on fine-grained manipulation, the overall policy only needs to learn the macro-level sequence, enabling stable performance over long-horizons. ChainedDiffuser (Xian et al., 2023) also uses a multilevel architecture where 3D observation data is first used to predict a macro-action, which indicates the next key position for the end-effector. Then, a low-level trajectory diffusion policy generates the local motion segments

that connect these macro-actions. Huang et al. (2024f, d) propose to generate sub-goals as waypoints guidance, helping the policy to break down long-horizon tasks.

Another aspect of long-horizon planning involves failure detection. Due to the diversity of application scenarios, policies often encounter out-of-distribution (OOD) problems in long-horizon tasks. Sentinel (Agia et al., 2024) categorizes failures into two types. For erratic failures that are urgent and need immediate correction, Sentinel checks the temporal consistency of the action sequence by measuring the overlap in its distribution. In contrast, for less urgent failures, the system queries a VLM to determine whether the current execution is failing to meet its intended progress.

4 Diffusion as Synthesizer

In the field of robotics, diffusion models are employed for more than just control policy. In addition to action-space generation such as action trajectory denoising, another line of methods leverage image and video diffusion generation models for robotic planning and data collection. In this section, we review current research on related approaches and discuss their benefits for robotic systems. Key methods are listed in Table 1.

4.1 Diffusion Generation Models for Planning

Owing to the powerful generative capabilities of contemporary diffusion-based image and video generators, many approaches have emerged that synthesize an "imagined" future execution process based on the current observations and instructions. The generated visual sequences can then serve as goals or execution paths that help to guide and refine the actual control policy.

Some early methods utilize diffusion-based image generation to rearrange objects within a scene. For instance, DALL-E-Bot (Kapelyukh et al., 2023) segments different objects in the initial observation image and then employs web-scale diffusion models to synthesize an ordered arrangement guided by a language prompt. The generated images provide the necessary poses for subsequent robot execution.

In order to guide robot policy execution, augmenting language instructions with image goal information provides a richer and more precise representation of the desired goal state. During real-world inference, many researchers propose incorporating diffusion generation models to synthesize goal images or execution videos that serve as visual guidance for the robot's control policy. CLOVER (Bu et al., 2024) employs a text-conditioned RGB-D video diffusion model to generate a sequence of visual sub-goals for execution, and then derives actions from these sub-goals while adaptively replanning in a closed-loop manner. This&That (Wang et al.,

2024a) introduces language-gesture conditioning to generate robot execution videos by explicitly specifying the manipulated objects and their targets positions. Dreamitate (Liang et al., 2024a) explores mapping human demonstrations videos to robot actions. It finetunes a video model to synthesize human execution videos where a person uses a customized tool that facilitates 3D tracking. During robot inference, the video model generates human execution videos, from which the tool trajectories can be easily extracted and applied to robot actions. AVDC (Ko et al., 2023) also trains a conditional diffusion model to synthesize an imagined execution video for the task, then computes the optical flow from the video and maps it to robot commands. Recent work including CoT-VLA (Zhao et al., 2025) incorporates chain-of-thought reasoning into VLA by predicting future goals, which serve as a guidance for action prediction. Other diffusion generator-based methods propose various insightful techniques, including 3D point clouds generation (Huang et al., 2024b), objects affordance flow generation (Tang et al., 2024) and image sub-goals generation (Zhen et al., 2024; Li et al., 2025; Pathre et al., 2024).

A recent trend in video generation is the exploration of diffusion world models (Bruce et al., 2024; Labs, 2024) that are trained on extensive data corpora and scaled to large architectures. These models are capable of modeling real-world physics and environments, exhibiting a robust understanding of the physical world, and are consequently applied in numerous robotics research. Some early explorations include UniPi (Du et al., 2023b) and UniSim (Yang et al., 2023), both of which train a text-conditioned video generation model on diverse sources of robotics data. These video models are capable of modeling a wide range of robotic tasks, serving as planners to facilitate real-robot execution. VLP (Du et al., 2023a) and HiP (Ajay et al., 2023) both propose a hierarchical policy. They combine video models with an LLM/VLM, using generated language plans to construct video plans that are subsequently used to condition low-level action policies. Moreover, VLP incorporates VLM models to assess the generated video plans through tree search. RoboDreamer (Zhou et al., 2024) decomposes language instructions into several components and introduces a novel method for composing the diffusion process on each language subcomponent, thereby enhancing the model's generation ability for new instructions. Additionally, some works explore robotic world models for navigation tasks (Shah et al., 2023) and trajectory-to-video tasks, such as IRASim (Zhu et al., 2024) and cosmos-predict (Agarwal et al., 2025). However, the significant computational overhead associated with these world models hinders their deployment in real-world applications that require efficient execution. Accelerating and improving the efficiency of video planners remains an unresolved direction for future research.

Table 1 Summary of robotics methods incorporating diffusion image and video generation models

Task	Taxonomy	Methods
Planning	Goal Generation	eg. CLOVER (Bu et al., 2024), Dreamitate (Liang et al., 2024a), AVDC (Ko et al., 2023), Imagination Policy (Huang et al., 2024b), Tang et al. (Tang et al., 2024), GR-MG (Li et al., 2025), Imagine2Servo (Pathre et al., 2024), 3D-VLA (Zhen et al., 2024)
	World Models	eg. UniPi (Du et al., 2023b), UniSim (Yang et al., 2023), VLP (Du et al., 2023a), HiP (Ajay et al., 2023), RoboDreamer (Zhou et al., 2024), IRASim (Zhu et al., 2024), ViNT (Shah et al., 2023), cosmos-predict (Agarwal et al., 2025)
Data Collection	Augmentation	eg. ROSIE (Yu et al., 2023), GenAug (Chen et al., 2023), CACTI (Mandi et al., 2022), DMD (Zhang et al., 2024b)

4.2 Diffusion Generation Models for Data Collection

Robotic learning constantly faces a major challenge, *i.e.*, the scarcity of data. It is widely held that the primary bottleneck in scaling powerful robotic models is the limited availability of high-quality, diverse datasets compared to those in other domains, such as computer vision or natural language processing. Early methods attempted to address this challenge by rendering simulated data to appear more realistic (Ho et al., 2021). However, the sim-to-real gap remains substantial. Recently, to tackle this issue, several approaches have proposed leveraging diffusion generation models for data collection or augmentation by synthesizing realistic videos of robot manipulation.

ROSIE (Yu et al., 2023) proposes using semantic image augmentation to scale up robot data. It leverages a diffusion text-guided image editing model to modify a specified region in accordance with the prompt. The augmentation text prompt may be hand-crafted or generated by an LLM, ensuring that the augmented videos lie outside the original data distribution. In the augmented data, the actions remain unchanged because the image editor only alters the objects without modifying the execution trajectory. Similarly, GenAug (Chen et al., 2023) and CACTI (Mandi et al., 2022) employ diffusion generation models to edit objects and environments or to add distractors, thereby achieving data augmentation. DMD (Zhang et al., 2024b) introduces generation models for collecting DAgger (Ross et al., 2011) samples. Specifically, it generates an augmentation dataset comprising eye-in-hand camera views captured from perspectives distinct from the original demonstrations, thus supplementing the data with out-of-distribution samples.

Most of current data augmentation frameworks involve only altering objects and environments in the original videos, but the action trajectories remain unchanged, thus limiting their scalability to more diverse data. A possible future exploration is to find world models with real physical understanding that can generate videos along with action data, which can be a promising way to scale up robot data.

5 Robotics Benchmarks and Datasets

In this section, we introduce several robotics benchmarks that are widely recognized and utilized by researchers. Due to the inherent complexity and variability of robotic tasks, standardizing all experimental settings in real-world experiments is challenging. Consequently, most benchmarks are developed within simulation environments.

Meta-World (Yu et al., 2020) is one of the earliest benchmarks focusing on multi-task robotics learning. It is an open-source simulated benchmark for meta-reinforcement learning, comprising 50 distinct robotic manipulation tasks, each crafted to provide a diverse, yet structured set of challenges. Meta-World aims to help evaluate algorithms' ability to generalize to entirely new, held-out tasks.

RLBench (James et al., 2020) is an early open-source learning environment tailored for advancing research in vision-guided robotic manipulation. It offers a large-scale dataset comprising 100 unique, hand-designed tasks of varying complexity. Notably, RLBench introduces a few-shot learning challenge to assess models' ability to generalize to new tasks by observing only a few demonstrations.

CALVIN (Mees et al., 2022) is an open-source simulated benchmark for long-horizon language-conditioned tasks. It provides an environment that features 4 different scenarios and 34 unique tasks, as well as a dataset with 40M trajectory data. A key contribution of CALVIN is its focus on long-horizon tasks, challenging agents to maintain coherence over extended action sequences.

Bi-DexHands (Chen et al., 2022) is an open-source simulated benchmark designed to advance research in bimanual dexterous manipulation. It offers a suite of tasks that simulate complex interactions requiring high degrees of freedom and coordination between two robotic hands.

DexArt (Bao et al., 2023) is an open-source simulated benchmark introducing multiple tasks for dexterous hands manipulating articulated objects. It comprises four tasks that assess a model's ability to generalize learned policies to unseen articulated objects. DexArt utilizes 3D point cloud data of objects as input for observations.

LIBERO (Liu et al., 2023) is an open-source simulated benchmark for lifelong robot manipulation learning, featuring 130 procedurally generated tasks across 4 suites. It focuses on transferring both declarative and procedural knowledge through language-conditioned tasks inspired by human activities. LIBERO provides high-quality demonstrations and evaluates key challenges like knowledge transfer, policy architectures, and task ordering robustness.

DROID (Khazatsky et al., 2024) is a large-scale, in-the-wild manipulation dataset comprising 76k demonstration trajectories of interaction data. The dataset spans 84 tasks and 564 distinct scenes, collected collaboratively by 50 collectors. DROID aims to enhance the generalization and robustness of robot manipulation policies by providing diverse, real-world manipulation data. The dataset, alongside policy learning code and a detailed hardware reproduction guide, is open-sourced to support reproducibility and further research.

BridgeData V2 (Ebert et al., 2021; Walke et al., 2023) is an extensive open-source real-world dataset comprises 60,096 trajectories of 13 skills collected across 24 diverse environments using a publicly available, low-cost robot. BridgeData V2 spans a variety of environments, enabling the development of skills that generalize across different settings, domains, and institutions.

Open X-Embodiment (O'Neill et al., 2023) is a comprehensive real-world dataset designed for general-purpose robotics learning across diverse robot types and tasks. It encompasses data from 22 distinct robot embodiments, including single-arm robots, bimanual systems, and quadrupeds. The dataset comprises over 1 million real-world robot trajectories, demonstrating 527 skills and 160,266 tasks, facilitating the training of generalist models capable of adapting to various embodiments and tasks.

RoboArena (Atreya et al., 2025) is a real-world benchmark for evaluating generalist robot policies at scale through a distributed evaluation network. Rather than fixing a single set of tasks or environments, RoboArena crowdsources double-blind pairwise comparisons across diverse real-world settings. It is instantiated across 7 academic institutions using the DROID robot platform, aiming to provide more scalable, trustworthy, and comparable evaluation for modern robot policies.

Current robotics research still lacks universally accepted real-world benchmarks, making it challenging to objectively compare the performance of different methods. Identifying unified, robust, and fair benchmarks for real-world settings remains an important direction for future work.

6 Task Objectives and Applications

In this section, we conclude the major task objectives and applications that diffusion-based methods typically address in robotics. Table 2 lists robotics methods with diffusion policy, each method labeled with its primary objective and methodology.

6.1 Constrained Motion Planning

Motion planning in a manipulation task requires the system to plan an action sequence that meets certain geometrical constraints, which is key to ensuring that the generated trajectories are safe. This typically means designing the planner to avoid collisions with obstacles in the environment. Recent work has started to look into the role of diffusion models in trajectory optimization. They modify the reverse diffusion process to align generated actions with the imposed constraints.

MPD (Carvalho et al., 2023) first proposes to use the diffusion model as constraint motion planning model. It trains the model on a set of collision-free trajectories so that it learns the posterior distribution of trajectories given the task objective. During inference, the diffusion model's denoising process works along with moving the trajectories to a low cost region defined by constraints. EDMP (Saha et al., 2024) traces a set of collision costs and optimizes along with the diffusion model. Similarly, Lan-o3dp (Li et al., 2024) computes the constraint cost by leveraging a language model. SafeDiffuser (Xiao et al., 2023) also injects invariance in diffusion dynamics to ensure safety during inference. Power et al. (Power et al., 2023) propose a method using composable diffusion models to generate trajectories that satisfy complex constraints, even for unseen combinations of constraints. The diffusion model is designed to compose multiple constraints at inference time, and the results are further refined via CSVTO (Power and Berenson, 2024).

Except for rigging the denoising process of diffusion models, various methods have been proposed to leverage diffusion models in constraint based motion planning. DiMSam (Fang et al., 2024) proposes to compose diffusion models through a task planner. The planner searches for the task plan, while learned diffusion samplers find continuous values satisfying the constraints. DiffusionSeeder (Huang et al., 2024c) uses the diffusion model as a trajectory seed generator, which speeds up the original collision-free optimization process.

6.2 Dexterous Manipulation

Biological evidence suggests that the dexterous structure of human hands facilitates the use of everyday tools. This inspired researchers to come up with dexterous manipulation

tasks, *i.e.*, adopting multiple manipulators or fingers to grasp and manipulate objects (Okamura et al., 2000). Some recent methods introduce diffusion policy into dexterous manipulation.

DexDiffuser (Weng et al., 2024) proposes DexSampler, which is conditioned on 3D object point cloud features and generates grasp poses via a diffusion process. The method then employs DexEvaluator to refine and rank the candidate grasps, executing the one with the highest score on the real robot. GraspGF (Wu et al., 2024b) primarily focuses on human-assisting dexterous grasping, leveraging a score-based generative model as a primitive policy and refining it further using a residual policy via reinforcement learning. DexGrasp-Diffusion (Zhang et al., 2024c) combines multi-dexterous hand grasp estimation with open-vocabulary affordance detection to better adapt to complex tasks. Other methods include ViViDex (Chen et al., 2024b) and Pan et al. (Pan et al., 2024), which use video demonstrations and VLAs, respectively, as conditioning modules for diffusion policy to facilitate dexterous manipulation. Overall, research on dexterous manipulation using diffusion policies is still in its preliminary stages, and we look forward to seeing amounts of work emerging in this area in the future.

6.3 Human-Robot Interaction

Human-Robot Interaction (HRI) enables seamless collaboration between humans and robots in dynamic environments, requiring intuitive communication, safety, and trust. Diffusion Co-policy (Ng et al., 2023) is the first to use diffusion models to learn a robot collaborative policy that can synergize with a human partner. At any timestep, the human can take action jointly with the robot action to complete collaborative tasks. Diffusion Co-policy proposes a transformer-based diffusion policy to predict future joint action sequences conditioned on past observations and human actions. Other methods take inspiration from reinforcement learning from human feedback (RLHF) (Christiano et al., 2017), explore learning policy from human feedback to shape the robot's behavior aligned with human preferences. For instance, representation-aligned preference-based learning (RAPL) (Tian et al., 2024) leverages human feedback to finetune the pretrained vision encoders so they match the end-user's visual representation. The aligned representations are then used as visual reward to guide the diffusion policy in following human intention.

Another perspective is to let robots learn from human video demonstrations, which significantly reduces the resources required for data gathering. Several approaches in dexterous manipulation (Qin et al., 2022; Chen et al., 2024b) use a pose estimation algorithm to convert human operating videos into 3D hand trajectories, and these representations are then used to learn control policies.

6.4 Mobile Manipulation

Despite recent progress in manipulation learning, mobile manipulation - an essential ability enabling robots to perform manipulation tasks in a larger space - remains challenging. The difficulty comes from dynamic environments that require the policy to have a strong understanding of the scene. Because of diffusion models' high-dimensional solution space and high precision, many methods start to leverage them in mobile manipulation.

Yang et al. (2024b) propose a universal policy that can control different robot embodiments, including a manipulator and a mobile base. They train a goal-conditioned large scale policy using a transformer block and a diffusion action head, demonstrating that the co-trained policy can achieve better performance on both navigation and manipulation tasks. UMI on Legs (Ha et al., 2024) uses a high level diffusion-based manipulation policy together with a low level whole body controller. The whole body controller is trained with RL to infer arm and leg actions in the world-frame rather than the body-frame. M²Diffuser (Yan et al., 2024) takes in robot-centric 3D scene visual input and uses an iterative denoising process to generate whole body control signals. During each iteration, the diffusion model's output is further refined by applying task objectives and physical constraints to reduce physical violations and execution errors.

6.5 Deformable Objects

Deformable objects manipulation is crucial for applications like surgery and manufacturing, where tasks may involve handling tissues, garments or clay. This remains a big challenge since modeling the interaction between the robot and deformable materials is difficult due to their complex and unpredictable physical behavior. Recent advances in diffusion policy, which has exhibited strong expressivity in other manipulation tasks, have prompted researchers to explore their integration into deformable objects manipulation.

Movement Primitive Diffusion (MPD) (Scheikl et al., 2024) focuses on gentle manipulation of deformable objects, primarily tissue and cloth. MPD combines the advantage of both score-based generative models and Probabilistic Dynamic Movement Primitives (ProDMPs) (Li et al., 2023). It leverages score-based denoising to produce ProDMP weight vectors, which are finally decoded into smooth, high-frequency trajectories with guaranteed initial conditions. This data-efficient, multimodal approach ensures gentle and precise manipulation for deformable objects. SculptDiff (Bartsch et al., 2024) studies clay sculpting in various shapes, extending into 3D deformable manipulation. SculptDiff proposes a diffusion policy conditioned on 3D point clouds representations of current and goal states, generating action sequences through iterative denoising. RoPotter (Yoo et al.,

Table 2 Overview of robotics methods with diffusion policy sorted by the publish date. The displayed methods are selected to ensure coverage of various methodologies and to include the most influential works from each subfields

Paper	Model	Algorithm	Objective	Methodology	Date
(Urain et al., 2023)	SE(3)-DiffusionFields	SGM			9/22
(Chi et al., 2023)	Diffusion Policy	DDPM/DDIM			3/23
(Reuss et al., 2023)	BESO	SGM			4/23
(Ha et al., 2023)	Ha et al.	DDPM/DDIM			7/23
(M. Xu et al., 2023)	XSkill	DDPM/DDIM			7/23
(Simeonov et al., 2023)	RPDiff	DDPM/DDIM			7/23
(Xian et al., 2023)	ChainedDiffuser	DDPM/DDIM			8/23
(Carvalho et al., 2023)	MPD	SGM			8/23
(Saha et al., 2024)	EDMP	DDPM/DDIM			9/23
(Sridhar, Shah, Glossop, & Levine, 2024)	NoMaD	DDPM/DDIM			10/23
(J. Liu, Stamatopoulou, & Kanoulas, 2024)	DiPPeR	DDPM/DDIM			10/23
(L. Chen et al., 2023)	PlayFusion	DDPM/DDIM			12/23
(Z. Liang et al., 2024)	SkillDiffuser	DDPM/DDIM			12/23
(Mishra et al., 2023)	GSC	DDPM/DDIM			1/24
(Weng et al., 2024)	DexDiffuser	DDPM/DDIM			2/24
(Ke et al., 2024)	3D Diffuser Actor	DDPM/DDIM			2/24
(Ze et al., 2024)	3D Diffuser Policy	DDPM/DDIM			3/24
(Z. Huang et al., 2024)	Subgoal Diffuser	DDPM/DDIM			3/24
(Z. Chen et al., 2024)	ViViDex	DDPM/DDIM			4/24
(C. Wang et al., 2024)	RISE	DDPM/DDIM			4/24
(Prasad et al., 2024)	Consistency Policy	Consistency Model			5/24
(W. Liu, Mao, et al., 2024)	CPM	DDPM/DDIM			5/24
(Bharadhwaj et al., 2024)	Track2Act	DDPM/DDIM			5/24
(Jia et al., 2024)	MaIL	DDPM/DDIM			6/24
(J. Yang, Cao, et al., 2024)	EquiBot	DDPM/DDIM			7/24
(Reuss et al., 2024)	MDT	SGM			7/24
(Wen et al., 2024)	TinyVLA	DDPM/DDIM			9/24
(Chisari et al., 2024)	PointFlowMatch	Flow Matching			9/24
(Rouxel et al., 2024)	Rouxel et al.	Flow Matching			10/24
(Y. Wang et al., 2024)	GenDP	DDPM/DDIM			10/24
(W. Liu, Wang, et al., 2024)	ForceMimic	DDPM/DDIM			10/24
(B. Huang et al., 2024)	3D ViTac	DDPM/DDIM			10/24
(S. Liu et al., 2024)	RDT-1B	DDPM/DDIM			10/24
(T. Chen et al., 2024)	G3Flow	DDPM/DDIM			11/24
(Hu et al., 2024)	VPP	DDPM/DDIM			12/24
(Xue et al., 2025)	RDP	DDPM/DDIM			3/25
(Zhao et al., 2025)	CoT-VLA	DDPM/DDIM			3/25
(Z. Hou et al., 2025)	Dita	DDPM/DDIM			3/25

: Policy Generalizability, : Long-Horizon Planning, : Constraint Planning, : Dexterous Manipulation, : Navigation, : System Efficiency, : Other Specialized Tasks

: Hierarchical Planning, : RGBD Perception, : 3D Perception, : Tactile Perception

: Image Goal Conditioned, : Language Goal Conditioned, : Object-centric Conditioned

2024) focuses on robot performing pottery wheel task, which also conditions the diffusion policy on a 3D point cloud observation of the clay shape. Overall, although much effort has been invested in deformable objects manipulation, how to effectively maximize diffusion policy's potential still remains an open question and represents a vast frontier.

6.6 Navigation

Robot navigation is another branch of robotic research that typically involves legged or wheeled robots moving in known or unknown environments. Some methods have employed diffusion models to capture multimodal action distributions without cost increases.

NoMaD (Sridhar et al., 2024) trains a diffusion policy that can utilize the task-agnostic behavior for exploration and switching to task-specific behavior as needed to solve the task. It introduces a goal-masking strategy to jointly encode the two behaviors in the observation context, which is then used to condition the diffusion policy to generate actions. DiPPeR (Liu et al., 2024a) focuses on maze navigation for legged robots. It conditions the trajectory path diffusion policy on a maze map image as well as on the start and goal coordinates. Similarly, LDP (Yu et al., 2024) employs a diffusion process for robot collision avoidance, incorporating global paths as additional guidance to facilitate the policy better navigate through maze like scenarios. DTG (Liang et al., 2024b) focuses on global navigation in large-scale environments by conditioning the diffusion model on LiDAR data, robot velocities, and target information to enhance its environmental understanding.

7 Challenges and Future Directions

The past several years have witnessed revolutionary progress brought by diffusion policies to robot learning. However, current state of technology and research remains far from perfect and widely applicable. In this section, we examine some of the key challenges faced by the field, as well as potential directions for promising future exploration.

7.1 Foundation Models

Foundation models have transformed the landscape of computer vision (CV) and natural language processing (NLP), where scaling up model capacity and data diversity has led to emergence ability on new tasks. Models like GPT-4 in NLP and ViTs in CV have shown how pretraining on massive internet-scale datasets enables downstream application. In robotics, the foundation models research is still in its early stages. Recent VLA models represent a key step toward generalist robotic agents by aggregating multiple modalities.

However, current VLAs often struggle with task accuracy and fine-grained control when compared to diffusion policy. Looking ahead, a promising direction is scaling up diffusion policy into foundation models. There have been some efforts: RDT-1B (Liu et al., 2024b) pretrains a transformer-based diffusion policy and scales it up to 1B parameters, demonstrating the potential of diffusion policies as generalist models. Dita (Hou et al., 2025) introduces a lightweight, 334 M-parameter DiT that employs in-context conditioning—directly denoising continuous action sequences from raw visual and language tokens, achieving strong zero- and few-shot generalization on long-horizon robotic tasks. Another promising avenue involves integrating diffusion policies with VLA architectures, thereby unifying the strengths of both paradigms, as several early methods have explored (Wen et al., 2024; Zhen et al., 2024; Black et al., 2024).

Despite these advancements, building robotic foundation models presents unique challenges. Unlike NLP and CV where large-scale datasets are abundant, robotics suffer from extreme data scarcity due to the cost and time of real-world data collection. Additionally, modeling the complex dynamics and multimodal interactions of the physical world remains difficult. Addressing these issues may require leveraging high-fidelity simulation environments and cross-domain data augmentation to scale data acquisition effectively. Furthermore, combining data-efficient algorithms with generalist architectures could also help enable scalable foundation models capable of robust generalization across diverse robotic tasks.

7.2 Multiple Modalities

Humans perceive and interact with the physical world through multiple sensory modalities—vision, touch, hearing, and more. It is natural to envision robotic systems equipped with similar multimodal capabilities to improve perception, reasoning, and control in dynamic environments. Recently, diffusion policies have been extended to tactile-based robotic tasks (George et al., 2024; Liu et al., 2024b; Huang et al., 2024a), demonstrating impressive potential in areas like accurate object manipulation and grasping. These methods often combine vision and tactile feedback to enhance robustness in tasks that require fine manipulation and dexterity.

However, integrating tactile information into diffusion policy still presents several open challenges. First, there is a lack of unified tactile sensing standards. Existing tactile sensors vary in data representation, resolution and noise characteristics, making it difficult to develop cross-embodiment generalizable models. Second, the fusion of tactile data with 3D point clouds remains underexplored. Efficiently aligning tactile feedback with 3D representations to infer object pose and contact locations is crucial for enabling more accurate and context-aware manipulation. Third, drawing inspiration

from human perception, tactile information can be crucial in situations where visual input is unavailable. Therefore, training models to perceive objects independently through vision and touch while building implicit connection is another promising direction.

7.3 Computation Efficiency

Real-time performance is a fundamental requirement for robotic systems, where policies are expected to generate actions with minimal latency. This imposes a critical demand for faster inference in diffusion policy-based methods, which are typically computationally intensive due to their iterative sampling processes. Recent advancements have sought to address this bottleneck, including leveraging of Mamba models (Cao et al., 2024; Jia et al., 2024), Consistency models (Prasad et al., 2024) and flow matching (Chisari et al., 2024; Rouxel et al., 2024).

Future directions could focus on integrating parallel and distributed computing strategies to further minimize latency, such as leveraging GPU/TPU acceleration and asynchronous sampling pipelines. Model compression techniques, including knowledge distillation and weight quantization, also present promising solutions for reducing model size without compromising performance. Furthermore, data efficiency remains underexplored in diffusion policies; designing lightweight, generalizable models capable of learning from fewer parameters and limited data could be useful for resource-constrained robotic systems.

7.4 Model Reliability

As real-world robotic applications increasingly rely on learned policies for complex and sensitive tasks, ensuring safety and transparency becomes important. Especially, real-world robotics often involves safety-critical operations such as surgical assistance, human-robot collaboration, and industrial manipulation, where any failure or unexpected behavior can have significant consequences. Therefore it is essential to develop policy models that are interpretable and reliable. Recently Chen et al. (2024a) proposes to attack the visual inputs in robot systems, successfully fooling the diffusion process into generating wrong actions. This highlights the vulnerability of diffusion policy to adversarial manipulations.

An important direction for future research is enhancing the interpretability and reliability of diffusion policy models. While many current methods utilizing language models as high level planners provide a degree of explainability at the planning stage, the underlying diffusion policy remains largely a black box. Future research should focus on improving the transparency of diffusion-based policies, potentially by integrating explainable AI techniques, uncertainty estima-

tion, or hybrid architectures that allow for self-reflection and reasoning over policy decisions.

7.5 World Models for Data Collection

Robotic learning has long been constrained by data scarcity, limiting the development of robust and generalizable policies. World models, particularly those based on diffusion video generation, are emerging as a promising solution by providing simulated environments for data collection and augmentation. These models aim to replace costly and time-consuming real-world data acquisition with high-fidelity simulations that can support policy learning at scale. Recent advancements including ROSIE (Yu et al., 2023), GenAug (Chen et al., 2023) have demonstrated effectiveness in generating diverse video sequences for data augmentation in robotic tasks.

However, current world models still show limited capabilities in accurately modeling realistic physical interactions and dynamics. Furthermore, these models predominantly focus on generating visual data in the form of videos without other crucial modalities such as trajectories, force feedback and tactile information that are essential for robot policy learning. Future research should investigate the development of better physics-aware world models that go beyond video generation to include comprehensive robotic data representations. By integrating these modalities, next-generation world models can better bridge the gap between simulation and reality, enabling more effective and data-efficient robot learning.

8 Conclusion

Robot learning has seen rapid progress in recent years, and diffusion models—known for their strong generative and expressive capabilities—are driving a new wave of advancements in the field. In this survey, we provide a comprehensive review of diffusion-based methods in robotics, covering both action-space generation through diffusion policies, and the use of image and video diffusion synthesizer to aid robot learning. Despite these advancements, developing universal and generalizable diffusion policies remains an open challenge. We have outlined several promising directions that may help address current limitations and push the field forward. Through this survey, we hope to offer a solid foundational resource and inspire researchers from both diffusion modeling and robotics backgrounds. We also encourage practitioners to explore and advance research in this exciting and rapidly evolving area.

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Data Availability No new datasets were generated in this study. The data supporting the findings of this survey are derived from publicly available sources and published literature, all of which are cited in the manuscript. Any curated summaries or analyses are based on these publicly accessible resources.

Declarations

Conflict of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Aburub, M., Beltran-Hernandez, C. C., Kamijo, T., & Hamaya, M. (2024). Learning diffusion policies from demonstrations for compliant contact-rich manipulation. [arXiv:2410.19235](#).
- Agarwal, N., Ali, A., Bala, M., Balaji, Y., Barker, E., Cai, T., Chattopadhyay, P., Chen, Y., Cui, Y., Ding, Y., et al. (2025). Cosmos world foundation model platform for physical AI. [arXiv:2501.03575](#)
- Agia, C., Sinha, R., Yang, J., Cao, Z.-A., Antonova, R., Pavone, M., Bohg, J. (2024). Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress. [arXiv preprint arXiv:2410.04640](#)
- Ajay, A., Du, Y., Gupta, A., Tenenbaum, J., Jaakkola, T., & Agrawal, P. (2022). Is conditional generative modeling all you need for decision-making? [arXiv:2211.15657](#)
- Ajay, A., Han, S., Du, Y., Li, S., Gupta, A., Jaakkola, T., & Agrawal, P. (2023). Compositional foundation models for hierarchical planning. *Advances in Neural Information Processing Systems*, 36, 22304–22325.
- Alterovitz, R., Koenig, S., & Likhachev, M. (2016). Robot planning in the real world: Research challenges and opportunities. *AI Magazine*, 37(2), 76–84.
- Ankile, L., Simeonov, A., Shenfeld, I., & Agrawal, P. (2024). Juicer: Data-efficient imitation learning for robotic assembly. [arXiv:2404.03729](#)
- Atreya, P., Pertsch, K., Lee, T., Kim, M.J., Jain, A., Kuramshin, A., Eppner, Clemens., Neary, C., Hu, E., Ramos, F., et al. (2025). Roboarena: Distributed real-world evaluation of generalist robot policies. [arXiv:2506.18123](#)
- Bao, C., Xu, H., Qin, Y., & Wang, X. (2023). Dexart: Benchmarking generalizable dexterous manipulation with articulated objects. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 21190–21200)
- Bartsch, A., Car, A., Avra, C., & Farimani, A. B. (2024). Sculptdiff: Learning robotic clay sculpting from humans with goal conditioned diffusion policy. [arXiv:2403.10401](#)
- Bharadhwaj, H., Mottaghi, R., Gupta, A., & Tulsiani, S. (2024). Track2act: Predicting point tracks from internet videos enables generalizable robot manipulation. In *European conference on computer vision* (pp. 306–324)
- Black, K., Brown, N., Driess, D., Esmail, A., Equi, M., Finn, C., Fusai, N., Groom, L., Hausman, K., Ichter, B., et al. (2024). π_0 : A vision-language-action flow model for general robot control. [arXiv:2410.24164](#)
- Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Chen, X., Choremanski, K., Ding, T., Driess, D., Dubey, A., Finn, C., et al. (2023). Rt-2: Vision-language-action models transfer web knowledge to robotic control. [arXiv:2307.15818](#)
- Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Dabis, J., Finn, C., Gopalakrishnan, K., Hausman, K., Herzog, A., Hsu, J., et al. (2022). Rt-1: Robotics transformer for real-world control at scale. [arXiv:2212.06817](#)
- Bruce, J., Dennis, M. D., Edwards, A., Parker-Holder, J., Shi, Y., Hughes, E., Lai, M., Mavalankar, A., Steigerwald, R., & Apps, C., et al. (2024). Generative interactive environments. In *Forty-first international conference on machine learning: Genie*
- Bu, Q., Zeng, J., Chen, L., Yang, Y., Zhou, G., Yan, J., & Li, H. (2024). Closed-loop visuomotor control with generative expectation for robotic manipulation. [arXiv:2409.09016](#)
- Cao, J., Zhang, Q., Sun, J., Wang, J., Cheng, H., Li, Y., Ma, J., Shao, Y., Zhao, W., Han, G., et al. (2024). Mamba policy: Towards efficient 3d diffusion policy with hybrid selective state models. [arXiv:2409.07163](#)
- Carvalho, J., Le, A. T., Baierl, M., Koert, D., & Peters, J. (2023). Motion planning diffusion: Learning and planning of robot motions with diffusion models. In *IEEE/RSJ international conference on intelligent robots and systems (IROS)* (pp. 1916–1923)
- Chang, H., Boyalakuntla, K., Liu, Y., Zhang, X., Schramm, L., & Boularias, A. (2024). Dap: Diffusion-based affordance prediction for multi-modality storage. In *IEEE/RSJ international conference on intelligent robots and systems (IROS)* (pp. 9476–9481)
- Chen, L., Bahl, S., & Pathak, D. (2023). Playfusion: Skill acquisition via diffusion from language-annotated play. In *Conference on robot learning* (pp. 2012–2029)
- Chen, Z., Kiami, S., Gupta, A., & Kumar, V. (2023). Genaug: Retargeting behaviors to unseen situations via generative augmentation. [arXiv:2302.06671](#)
- Chen, T., Mu, Y., Liang, Z., Chen, Z., Peng, S., Chen, Q., Xu, M., Hu, R., Zhang, H., Li, X., et al. (2024). G3flow: Generative 3d semantic flow for pose-aware and generalizable object manipulation. [arXiv:2411.18369](#)
- Chen, Y., Xue, H., Chen, Y. (2024). Diffusion policy attacker: Crafting adversarial attacks for diffusion-based policies. [arXiv:2405.19424](#)
- Chen, Z., Chen, S., Arlaud, E., Laptev, I., & Schmid, C. (2024). Vividex: Learning vision-based dexterous manipulation from human videos [arXiv:2404.15709](#).
- Chen, Y., Wu, T., Wang, S., Feng, X., Jiang, J., Lu, Z., & Yang, Y. (2022). Towards human-level bimanual dexterous manipulation with reinforcement learning. *Advances in Neural Information Processing Systems*, 35, 5150–5163.

- Chisari, E., Heppert, N., Argus, M., Welschehold, T., Brox, T., & Valada, A. (2024). Learning robotic manipulation policies from point clouds with conditional flow matching. [arXiv:2409.07343](https://arxiv.org/abs/2409.07343)
- Chi, C., Xu, Z., Feng, S., Cousineau, E., Du, Y., Burchfiel, B., & Song, S. (2023). Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*. <https://doi.org/10.1177/02783649241273668>
- Christiano, P. F., Leike, J., Brown, T., Martic, M., Legg, S., & Amodei, D. (2017). Deep reinforcement learning from human preferences. In *Advances in neural information processing systems* (p. 30)
- Ding, H., Jaquier, N., Peters, J., & Rozo, L. (2024). Fast and robust visuomotor riemannian flow matching policy. [arXiv:2412.10855](https://arxiv.org/abs/2412.10855)
- Du, Y., & Mordatch, I. (2019). Implicit generation and modeling with energy based models. In *Advances in neural information processing systems* (p. 32)
- Du, Y., Yang, M., Florence, P., Xia, F., Wahid, A., Ichter, B., Sermanet, P., Yu, T., Abbeel, P., Tenenbaum, J.B., et al. (2023). Video language planning. [arXiv:2310.10625](https://arxiv.org/abs/2310.10625)
- Du, Y., Yang, S., Dai, B., Dai, H., Nachum, O., Tenenbaum, J., & Abbeel, P. (2023). Learning universal policies via text-guided video generation. *Advances in Neural Information Processing Systems*, 36, 9156–9172.
- Ebert, F., Yang, Y., Schmeckpeper, K., Bucher, B., Georgakis, G., Daniilidis, K., & Levine, S. (2021). Bridge data: Boosting generalization of robotic skills with cross-domain datasets. [arXiv:2109.13396](https://arxiv.org/abs/2109.13396).
- Fang, X., Garrett, C. R., Eppner, C., Lozano-Pérez, T., Kaelbling, L. P., & Fox, D. (2024). Dimsam: Diffusion models as samplers for task and motion planning under partial observability. In *IEEE/RSJ international conference on intelligent robots and systems (iros)* (pp. 1412–1419)
- Funk, N., Urain, J., Carvalho, J., Prasad, V., Chalvatzaki, G., & Peters, J. (2024). Actionflow: Equivariant, accurate, and efficient policies with spatially symmetric flow matching. [arXiv:2409.04576](https://arxiv.org/abs/2409.04576)
- George, A., Gano, S., Katragadda, P., & Farimani, A. B. (2024). Vital pretraining: Visuo-tactile pretraining for tactile and non-tactile manipulation policies. [arXiv:2403.11898](https://arxiv.org/abs/2403.11898)
- Gode, S., Nayak, A., & Burgard, W. (2024). Flownav: Learning efficient navigation policies via conditional flow matching. [arXiv:2411.09524](https://arxiv.org/abs/2411.09524)
- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., & Bengio, Y. (2014). Generative adversarial nets. In *Advances in neural information processing systems* (p. 27)
- Gu, A., & Dao, T. (2023). Mamba: Linear-time sequence modeling with selective state spaces. [arXiv:2312.00752](https://arxiv.org/abs/2312.00752)
- Gui, J., Sun, Z., Wen, Y., Tao, D., & Ye, J. (2021). A review on generative adversarial networks: Algorithms, theory, and applications. *IEEE Transactions on Knowledge and Data Engineering*, 35(4), 3313–3332.
- Guo, S. W., Hsiao, T. C., Liu, Y. L., & Lee, C. Y. (2024). Precise pick-and-place using score-based diffusion networks. In *IEEE/RSJ international conference on intelligent robots and systems (IROS)* (pp. 7484–7491)
- Ha, H., Florence, P., & Song, S. (2023). Scaling up and distilling down: Language-guided robot skill acquisition. In *Conference on robot learning* (pp. 3766–3777)
- Ha, H., Gao, Y., Fu, Z., Tan, J., & Song, S. (2024). Umi on legs: Making manipulation policies mobile with manipulation-centric whole-body controllers. [arXiv:2407.10353](https://arxiv.org/abs/2407.10353)
- Hansen-Estruch, P., Kostrikov, I., Janner, M., Kuba, J. G., & Levine, S. (2023). Idql: Implicit q-learning as an actor-critic method with diffusion policies. [arXiv:2304.10573](https://arxiv.org/abs/2304.10573)
- Hao, C., Lin, K., Luo, S., & Soh, H. (2024). Language-guided manipulation with diffusion policies and constrained inpainting. [arXiv:2406.09767](https://arxiv.org/abs/2406.09767)
- Hatch, K.B., Balakrishna, A., Mees, O., Nair, S., Park, S., Wulfe, B., Itkina, M., Eysenbach, B., Levine, S., Kollar, T., et al. (2024). Ghil-glue: Hierarchical control with filtered subgoal images. [arXiv:2410.20018](https://arxiv.org/abs/2410.20018)
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770–778)
- Ho, D., Rao, K., Xu, Z., Jang, E., Khansari, M., & Bai, Y. (2021). Retinagan: An object-aware approach to sim-to-real transfer. In *2021 IEEE international conference on robotics and automation (ICRA)* (pp. 10920–10926)
- Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33, 6840–6851.
- Ho, J., Salimans, T., Gritsenko, A., Chan, W., Norouzi, M., & Fleet, D. J. (2022). Video diffusion models. *Advances in Neural Information Processing Systems*, 35, 8633–8646.
- Hou, Z., Gao, M., Yu, H., Yang, M., & Jeong, C. I. (2024). Sdp: Spiking diffusion policy for robotic manipulation with learnable channel-wise membrane thresholds. [arXiv:2409.11195](https://arxiv.org/abs/2409.11195)
- Hou, Y., Liu, Z., Chi, C., Cousineau, E., Kuppuswamy, N., Feng, S., & Song, S. (2024). Adaptive compliance policy: Learning approximate compliance for diffusion guided control. [arXiv:2410.09309](https://arxiv.org/abs/2410.09309)
- Hou, Z., Zhang, T., Xiong, Y., Duan, H., Pu, H., Tong, R., Zhao, C., Zhu, X., Qiao, Y., Dai, J., et al. (2025). Dita: Scaling diffusion transformer for generalist vision-language-action policy. [arXiv:2503.19757](https://arxiv.org/abs/2503.19757)
- Hsu, C. C., Wen, B., Xu, J., Narang, Y., Wang, X., Zhu, Y., & Birchfield, S. (2024). Spot: Se (3) pose trajectory diffusion for object-centric manipulation. [arXiv:2411.00965](https://arxiv.org/abs/2411.00965)
- Hu, Y., Guo, Y., Wang, P., Chen, X., Wang, Y. J., Zhang, J., Chen, J. (2024). Video prediction policy: A generalist robot policy with predictive visual representations. [arXiv:2412.14803](https://arxiv.org/abs/2412.14803)
- Hu, Y., Lin, F., Zhang, T., Yi, L., & Gao, Y. (2023). Look before you leap: Unveiling the power of gpt-4v in robotic vision-language planning. [arXiv:2311.17842](https://arxiv.org/abs/2311.17842)
- Huang, T., Jiang, G., Ze, Y., & Xu, H. (2024). Diffusion reward: Learning rewards via conditional video diffusion. In *European conference on computer vision* (pp. 478–495)
- Huang, Z., Lin, Y., Yang, F., & Berenson, D. (2024). Subgoal diffuser: Coarse-to-fine subgoal generation to guide model predictive control for robot manipulation. [arXiv:2403.13085](https://arxiv.org/abs/2403.13085)
- Huang, H., Schmeckpeper, K., Wang, D., Biza, O., Qian, Y., Liu, H., & Walters, R. (2024). Imagination policy: Using generative point cloud models for learning manipulation policies. [arXiv:2406.11740](https://arxiv.org/abs/2406.11740)
- Huang, H., Sundaralingam, B., Mousavian, A., Murali, A., Goldberg, K., & Fox, D. (2024). Diffusionseeder: Seeding motion optimization with diffusion for rapid motion planning. [arXiv:2410.16727](https://arxiv.org/abs/2410.16727)
- Huang, B., Wang, Y., Yang, X., Luo, Y., & Li, Y. (2024). 3d-vitac: Learning fine-grained manipulation with visuo-tactile sensing. [arXiv:2410.24091](https://arxiv.org/abs/2410.24091)
- Huang, R., Liu, S., Pei, Y., Wang, P., Wang, G., Yang, Y., & Shen, H. (2024). Goal-reaching policy learning from non-expert observations via effective subgoal guidance. [arXiv:2409.03996](https://arxiv.org/abs/2409.03996)
- Hua, J., Zeng, L., Li, G., & Ju, Z. (2021). Learning for a robot: Deep reinforcement learning, imitation learning, transfer learning. *Sensors*, 21(4), 1278.
- Ikeda, T., Zakharov, S., Ko, T., Irshad, M. Z., Lee, R., Liu, K., & Nishiwaki, K. (2024). Diffusionnocs: Managing symmetry and uncertainty in sim2real multi-modal category-level pose estimation. In *IEEE/RSJ international conference on intelligent robots and systems (IROS)* (pp. 7406–7413)
- James, S., Ma, Z., Arrojo, D. R., & Davison, A. J. (2020). Rlbench: The robot learning benchmark & learning environment. *IEEE Robotics and Automation Letters*, 5(2), 3019–3026.

- Janner, M., Du, Y., Tenenbaum, J. B., & Levine, S. (2022). Planning with diffusion for flexible behavior synthesis. [arXiv:2205.09991](#)
- Jia, X., Wang, Q., Donat, A., Xing, B., Li, G., Zhou, H., & Neumann, G. (2024). Mail: Improving imitation learning with mamba. [arXiv:2406.08234](#)
- Kaelbling, L. P., Littman, M. L., & Moore, A. W. (1996). Reinforcement learning: A survey. *Journal of Artificial Intelligence Research*, 4, 237–285.
- Kalashnikov, D., Irpan, A., Pastor, P., Ibarz, J., Herzog, A., Jang, E., Quillen, D., Holly, E., Kalakrishnan, M., Vanhoucke, V., and others (2018). Scalable deep reinforcement learning for vision-based robotic manipulation. In *Conference on robot learning* (pp. 651–673).
- Kapelyukh, I., Vosylius, V., & Johns, E. (2023). Dall-e-bot: Introducing web-scale diffusion models to robotics. *IEEE Robotics and Automation Letters*, 8(7), 3956–3963.
- Ke, T. W., Gkanatsios, N., & Fragkiadaki, K. (2024). 3d diffuser actor: Policy diffusion with 3d scene representations. [arXiv:2402.10885](#)
- Khazatsky, A., Pertsch, K., Nair, S., Balakrishna, A., Dasari, S., Karamcheti, S., Nasiriany, S., Srirama, M. K., Chen, L. Y., & Ellis, K. (2024). Droid: A large-scale in-the-wild robot manipulation dataset. [arXiv:2403.12945](#)
- Kim, S., Choi, Y., Matsunaga, D. E., Kim, K.-E. (2024). Stitching sub-trajectories with conditional diffusion model for goal-conditioned offline rl. In *Proceedings of the AAAI conference on artificial intelligence* (Vol. 38, pp. 13160–13167).
- Kim, M. J., Finn, C., & Liang, P. (2025). Fine-tuning vision-language-action models: Optimizing speed and success. [arXiv:2502.19645](#)
- Kim, M.J., Pertsch, K., Karamcheti, S., Xiao, T., Balakrishna, A., Nair, S., Rafailov, R., Foster, E., Lam, G., Sanketi, P., et al. (2024). Openvla: An open-source vision-language-action model. [arXiv:2406.09246](#)
- Kingma, D. P., Welling, M., et al. (2013). *Auto-encoding variational bayes*. Canada: Banff.
- Ko, P. C., Mao, J., Du, Y., Sun, S. H., & Tenenbaum, J. B. (2023). Learning to act from actionless videos through dense correspondences. [arXiv:2310.08576](#)
- Kong, Z., Ping, W., Huang, J., Zhao, K., & Catanzaro, B. (2020). Diffwave: A versatile diffusion model for audio synthesis. [arXiv:2009.09761](#)
- Labs, W. (2024). *Generating worlds*. <https://www.worldlabs.ai/blog>
- LaValle, S. (1998). Rapidly-exploring random trees: A new tool for path planning. Research Report 9811
- Lehtomaki, N., Sandell, N., & Athans, M. (1981). Robustness results in linear-quadratic gaussian based multivariable control designs. *IEEE Transactions on Automatic Control*, 26(1), 75–93. <https://doi.org/10.1109/TAC.1981.1102565>
- Liang, J., Liu, R., Ozguroglu, E., Sudhakar, S., Dave, A., Tokmakov, P., & Vondrick, C. (2024). Dreamitate: Real-world visuomotor policy learning via video generation. [arXiv:2406.16862](#)
- Liang, Z., Mu, Y., Ma, H., Tomizuka, M., Ding, M., & Luo, P. (2024). Skilldiffuser: Interpretable hierarchical planning via skill abstractions in diffusion-based task execution. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16467–16476)
- Liang, J., Payandeh, A., Song, D., Xiao, X., & Manocha, D. (2024). Dtg: Diffusion-based trajectory generation for mapless global navigation. In *IEEE/RSJ international conference on intelligent robots and systems (IROS)* (pp. 5340–5347)
- Li, H., Feng, Q., Zheng, Z., Feng, J., & Knoll, A. (2024). Language-guided object-centric diffusion policy for collision-aware robotic manipulation. [arXiv:2407.00451](#)
- Li, G., Jin, Z., Volpp, M., Otto, F., Lioutikov, R., & Neumann, G. (2023). Prodm: A unified perspective on dynamic and probabilistic movement primitives. *IEEE Robotics and Automation Letters*, 8(4), 2325–2332.
- Lin, T., Zhang, Y., Li, Q., Qi, H., Yi, B., Levine, S., & Malik, J. (2024). Learning visuotactile skills with two multifingered hands. [arXiv:2404.16823](#)
- Liu, W., Du, Y., Hermans, T., Chernova, S., & Paxton, C. (2022). Struct-diffusion: Language-guided creation of physically-valid structures using unseen objects. [arXiv:2211.04604](#)
- Liu, W., Mao, J., Hsu, J., Hermans, T., Garg, A., & Wu, J. (2024). Composable part-based manipulation. [arXiv:2405.05876](#)
- Liu, J., Stamatopoulou, M., & Kanoulas, D. (2024). Dipper: Diffusion-based 2d path planner applied on legged robots. In *2024 IEEE international conference on robotics and automation (ICRA)* (pp. 9264–9270)
- Liu, S., Wu, L., Li, B., Tan, H., Chen, H., Wang, Z., & Zhu, J. (2024). Rdt-1b: a diffusion foundation model for bimanual manipulation. [arXiv:2410.07864](#)
- Liu, X., Gong, C., & Liu, Q. (2022). Flow straight and fast: Learning to generate and transfer data with rectified flow. [arXiv:2209.03003](#)
- Liu, W., Wang, J., Wang, Y., Wang, W., & Lu, C. (2024). Forcemimic: Force-centric imitation learning with force-motion capture system for contact-rich manipulation. [arXiv:2410.07554](#)
- Liu, B., Zhu, Y., Gao, C., Feng, Y., Liu, Q., Zhu, Y., & Stone, P. (2023). Libero: Benchmarking knowledge transfer for lifelong robot learning. *Advances in Neural Information Processing Systems*, 36, 44776–44791.
- Li, P., Wu, H., Huang, Y., Cheang, C., Wang, L., & Kong, T. (2025). Gr-mg: Leveraging partially-annotated data via multi-modal goal-conditioned policy. *IEEE Robotics and Automation Letters*. <https://doi.org/10.1109/LRA.2025.3526436>
- Lu, G., Gao, Z., Chen, T., Dai, W., Wang, Z., & Tang, Y. (2024). Manicm: Real-time 3d diffusion policy via consistency model for robotic manipulation. [arXiv:2406.01586](#)
- Lugmayr, A., Danelljan, M., Romero, A., Yu, F., Timofte, R., & Van Gool, L. (2022). Repaint: Inpainting using denoising diffusion probabilistic models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 11461–11471)
- Ma, X., Patidar, S., Haughton, I., & James, S. (2024). Hierarchical diffusion policy for kinematics-aware multi-task robotic manipulation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 18081–18090)
- Mandi, Z., Bharadhwaj, H., Moens, V., Song, S., Rajeswaran, A., & Kumar, V. (2022). Cacti: A framework for scalable multi-task multi-scene visual imitation learning. [arXiv:2212.05711](#)
- Mees, O., Hermann, L., Rosete-Beas, E., & Burgard, W. (2022). Calvin: A benchmark for language-conditioned policy learning for long-horizon robot manipulation tasks. *IEEE Robotics and Automation Letters*, 7(3), 7327–7334.
- Mishra, U. A., Xue, S., Chen, Y., & Xu, D. (2023). Generative skill chaining: Long-horizon skill planning with diffusion models. In *Conference on robot learning* (pp. 2905–2925)
- Mishra, U. A., C., & Y. (2024). Reorientdiff: Diffusion model based reorientation for object manipulation. In *2024 IEEE international conference on robotics and automation (ICRA)* (pp. 10867–10873)
- Nair, S., Rajeswaran, A., Kumar, V., Finn, C., & Gupta, A. (2022). R3m: A universal visual representation for robot manipulation. [arXiv:2203.12601](#)
- Ng, E., Liu, Z., & Kennedy, M. (2023). Diffusion co-policy for synergistic human-robot collaborative tasks. *IEEE Robotics and Automation Letters*. <https://doi.org/10.1109/LRA.2023.3330663>
- Okamura, A. M., Smaby, N., & Cutkosky, M. R. (2000). An overview of dexterous manipulation. In *Proceedings 2000 icra. millennium conference. IEEE international conference on robotics and automation. symposia proceedings (cat. no. 00ch37065)* (Vol. 1, pp. 255–262)
- O'Neill, A., Rehman, A., Gupta, A., Maddukuri, A., Gupta, A., Padalkar, A., Lee, A., Pooley, A., Gupta, A., Mandlekar, A., et al.

- (2023). Open x-embodiment: Robotic learning datasets and rt-x models. [arXiv:2310.08864](https://arxiv.org/abs/2310.08864)
- Pan, C., Junge, K., & Hughes, J. (2024). Vision-language-action model and diffusion policy switching enables dexterous control of an anthropomorphic hand. [arXiv:2410.14022](https://arxiv.org/abs/2410.14022)
- Pathre, P., Gupta, G., Qureshi, M. N., Brunda, M., Brahmabhatt, S., & Krishna, K. M. (2024). Imagine2servo: Intelligent visual servoing with diffusion-driven goal generation for robotic tasks. In *IEEE/RSJ international conference on intelligent robots and systems (IROS)* (pp. 13466–13472)
- Peebles, W., & Xie, S. (2023). Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 4195–4205)
- Perez, E., Strub, F., De Vries, H., Dumoulin, V., & Courville, A. (2018). Film: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI conference on artificial intelligence* (Vol. 32)
- Pomerleau, D. A. (1988). Alvin: An autonomous land vehicle in a neural network. *Advances in neural information processing systems* (Vol. 1)
- Power, T., Soltani-Zarrin, R., Iba, S., & Berenson, D. (2023). Sampling constrained trajectories using composable diffusion models. In *Iros 2023 workshop on differentiable probabilistic robotics: Emerging perspectives on robot learning*
- Power, T., & Berenson, D. (2024). Constrained stein variational trajectory optimization. *IEEE Transactions on Robotics*. <https://doi.org/10.1109/TRO.2024.3428428>
- Prasad, A., Lin, K., Wu, J., Zhou, L., & Bohg, J. (2024). Consistency policy: Accelerated visuomotor policies via consistency distillation. [arXiv:2405.07503](https://arxiv.org/abs/2405.07503)
- Qin, Y., Wu, Y. H., Liu, S., Jiang, H., Yang, R., Fu, Y., & Wang, X. (2022). Dexmv: Imitation learning for dexterous manipulation from human videos. *IN European conference on computer vision* (pp. 570–587)
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. (2021). Learning transferable visual models from natural language supervision. In *International conference on machine learning* (pp. 8748–8763).
- Rana, K., Abou-Chakra, J., Garg, S., Lee, R., Reid, I., & Suenderhauf, N. (2024). Affordance-centric policy learning: Sample efficient and generalisable robot policy learning using affordance-centric task frames. [arXiv:2410.12124](https://arxiv.org/abs/2410.12124)
- Ren, Y., Cong, Y., Chen, R., & Long, J. (2024). Learning generalizable 3d manipulation with 10 demonstrations. [arXiv:2411.10203](https://arxiv.org/abs/2411.10203)
- Ren, A. Z., Lidard, J., Ankile, L. L., Simeonov, A., Agrawal, P., Majumdar, A., & Simchowitz, M. (2024). Diffusion policy policy optimization. [arXiv:2409.00588](https://arxiv.org/abs/2409.00588)
- Reuss, M., Li, M., Jia, X., Lioutikov, R. (2023). Goal-conditioned imitation learning using score-based diffusion policies. [arXiv:2304.02532](https://arxiv.org/abs/2304.02532)
- Reuss, M., Yağmurlu, Ö. E., Wenzel, F., & Lioutikov, R. (2024). Multimodal diffusion transformer: Learning versatile behavior from multimodal goals. [arXiv:2407.05996](https://arxiv.org/abs/2407.05996)
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2021). High-resolution image synthesis with latent diffusion models
- Ross, S., & Bagnell, D. (2010). Efficient reductions for imitation learning. In *Proceedings of the thirteenth international conference on artificial intelligence and statistics* (pp. 661–668)
- Ross, S., Gordon, G., & Bagnell, D. (2011). A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the fourteenth international conference on artificial intelligence and statistics* (pp. 627–635)
- Rouxel, Q., Ferrari, A., Ivaldi, S., & Mouret, J. B. (2024). Flow matching imitation learning for multi-support manipulation. In *IEEE-RAS 23rd international conference on humanoid robots (humanoids)* (pp. 528–535)
- Ryu, H., Kim, J., An, H., Chang, J., Seo, J., Kim, T., Horowitz, R. (2024). Diffusion-edfs: Bi-equivariant denoising generative modeling on se (3) for visual robotic manipulation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 18007–18018).
- Saha, K., Mandadi, V., Reddy, J., Srikanth, A., Agarwal, A., Sen, B., & Krishna, M. (2024). Edmp: Ensemble-of-costs-guided diffusion for motion planning. In *2024 IEEE international conference on robotics and automation (ICRA)* (pp. 10351–10358)
- Scheikl, P. M., Schreiber, N., Haas, C., Freymuth, N., Neumann, G., Lioutikov, R., & Mathis-Ullrich, F. (2024). Movement primitive diffusion: Learning gentle robotic manipulation of deformable objects. *IEEE Robotics and Automation Letters*. <https://doi.org/10.1109/LRA.2024.3382529>
- Shah, D., Sridhar, A., Dashora, N., Stachowicz, K., Black, K., Hirose, N., & Levine, S. (2023). Vint: A foundation model for visual navigation. [arXiv:2306.14846](https://arxiv.org/abs/2306.14846)
- Simeonov, A., Goyal, A., Manuelli, L., Yen-Chen, L., Sarmiento, A., Rodriguez, A., & Fox, D. (2023). Shelving, stacking, hanging: Relational pose diffusion for multi-modal rearrangement. [arXiv:2307.04751](https://arxiv.org/abs/2307.04751)
- Song, Y., Dhariwal, P., Chen, M., & Sutskever, I. (2023). Consistency models. [arXiv:2303.01469](https://arxiv.org/abs/2303.01469)
- Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., & Poole, B. (2020). Score-based generative modeling through stochastic differential equations. [arXiv:2011.13456](https://arxiv.org/abs/2011.13456)
- Sridhar, A., Shah, D., Glossop, C., & Levine, S. (2024). Nomad: Goal masked diffusion policies for navigation and exploration. In *2024 IEEE international conference on robotics and automation (ICRA)* (pp. 63–70)
- Su, Y., Zhan, X., Fang, H., Li, Y. L., Lu, C., & Yang, L. (2024). Motion before action: Diffusing object motion as manipulation condition. [arXiv:2411.09658](https://arxiv.org/abs/2411.09658)
- Tang, W., Pan, J. H., Zhan, W., Zhou, J., Yao, H., Liu, Y. H., & Fu, C. W. (2024). Embodiment-agnostic action planning via object-part scene flow. [arXiv:2409.10032](https://arxiv.org/abs/2409.10032)
- Team, O.M., Ghosh, D., Walke, H., Pertsch, K., Black, K., Mees, O., Dasari, S., Hejna, J., Kreiman, T., Xu, C., et al. (2024). Octo: An open-source generalist robot policy. [arXiv:2405.12213](https://arxiv.org/abs/2405.12213)
- Tian, R., Wu, Y., Xu, C., Tomizuka, M., Malik, J., & Bajcsy, A. (2024). Maximizing alignment with minimal feedback: Efficiently learning rewards for visuomotor robot policy alignment. [arXiv:2412.04835](https://arxiv.org/abs/2412.04835)
- Urain, J., Funk, N., Peters, J., & Chalvatzaki, G. (2023). Se (3)-diffusionfields: Learning smooth cost functions for joint grasp and motion optimization through diffusion. In *2023 IEEE international conference on robotics and automation (ICRA)* (pp. 5923–5930)
- Van Den Oord, A., & Vinyals, O. (2017). Neural discrete representation learning. In *Advances in neural information processing systems* (p. 30)
- Walke, H.R., Black, K., Zhao, T.Z., Vuong, Q., Zheng, C., Hansen-Estruch, P., He, A.W., Myers, V., Kim, M.J., Du, M., et al. (2023). Bridgedata v2: A dataset for robot learning at scale. In *Conference on robot learning* (pp. 1723–1736).
- Wang, C., Fang, H., Fang, H. S., Lu, C. (2024). Rise: 3d perception makes real-world robot imitation simple and effective. [arXiv:2404.12281](https://arxiv.org/abs/2404.12281)
- Wang, D., Hart, S., Surovik, D., Kelestemur, T., Huang, H., Zhao, H., & Platt, R. (2024). Equivariant diffusion policy. [arXiv:2407.01812](https://arxiv.org/abs/2407.01812)
- Wang, D., Liu, C., Chang, F., & Xu, Y. (2024). Hierarchical diffusion policy: manipulation trajectory generation via contact guidance. [arXiv:2411.12982](https://arxiv.org/abs/2411.12982)
- Wang, B., Sridhar, N., Feng, C., Van der Merwe, M., Fishman, A., Fazeli, N., Park, J. J. (2024). This&that: Language-gesture controlled video generation for robot planning. [arXiv:2407.05530](https://arxiv.org/abs/2407.05530)

- Wang, Y., Yin, G., Huang, B., Kelestemur, T., Wang, J., Li, Y. (2024). Gendp: 3d semantic fields for category-level generalizable diffusion policy. In *8th annual conference on robot learning* (Vol. 2)
- Wen, J., Zhu, Y., Li, J., Zhu, M., Wu, K., Xu, Z., Liu, N., Cheng, R., Shen, C., Peng, Y., et al. (2024). Tinyvla: Towards fast, data-efficient vision-language-action models for robotic manipulation. [arXiv:2409.12514](#)
- Weng, Z., Lu, H., Kragic, D., & Lundell, J. (2024). Dexdiffuser: Generating dexterous grasps with diffusion models. [arXiv:2402.02989](#)
- Wolf, R., Shi, Y., Liu, S., & Rayyes, R. (2025). Diffusion models for robotic manipulation: A survey. *Frontiers in Robotics and AI*, 12, 1606247.
- Wu, T., Wu, M., Zhang, J., Gan, Y., Dong, H. (2024). Learning score-based grasping primitive for human-assisting dexterous grasping. In *Advances in Neural Information Processing Systems*, 36,
- Wu, Y., Chen, Z., Wu, F., Chen, L., Zhang, L., Bing, Z., & Haddadin, S. (2024). Tacdiffusion: Force-domain diffusion policy for precise tactile manipulation. [arXiv:2409.11047](#)
- Wu, T., Li, J., Zhang, J., Wu, M., & Dong, H. (2024). Canonical representation and force-based pretraining of 3d tactile for dexterous visuo-tactile policy learning. [arXiv:2409.17549](#)
- Xia, S., Fang, H., Lu, C., Fang, H.-S. (2024). Cage: Causal attention enables data-efficient generalizable robotic manipulation. [arXiv:2410.14974](#)
- Xian, Z., Gkanatsios, N., Gervet, T., Ke, T. W., & Fragkiadaki, K. (2023). *Chaineddiffuser: Unifying trajectory diffusion and key-pose prediction for robotic manipulation 7th annual conference on robot learning*.
- Xiao, W., Wang, T. H., Gan, C., & Rus, D. (2023). Safediffuser: Safe planning with diffusion probabilistic models. [arXiv:2306.00148](#)
- Xu, M., Xu, Z., Chi, C., Veloso, M., Song, S. (2023). Xskill: Cross embodiment skill discovery. In *Conference on robot learning* (pp. 3536–3555).
- Xue, H., Ren, J., Chen, W., Zhang, G., Fang, Y., Gu, G., & Lu, C. (2025). Reactive diffusion policy: Slow-fast visual-tactile policy learning for contact-rich manipulation. [arXiv:2503.02881](#)
- Xu, S., Wang, Y., Xia, C., Zhu, D., Huang, T., & Xu, C. (2025). Vla-cache: Towards efficient vision-language-action model via adaptive token caching in robotic manipulation. [arXiv:2502.02175](#)
- Xu, M., Yu, L., Song, Y., Shi, C., Ermon, S., & Tang, J. (2022). Geodiff: A geometric diffusion model for molecular conformation generation. [arXiv:2203.02923](#)
- Yan, S., Zhang, Z., Han, M., Wang, Z., Xie, Q., Li, Z., & Zhu, S. C. (2024). M2diffuser: Diffusion-based trajectory optimization for mobile manipulation in 3d scenes. [arXiv:2410.11402](#).
- Yang, M., Du, Y., Ghasemipour, K., Tompson, J., Schuurmans, D., Abbeel, P. (2023). Learning interactive real-world simulators. [arXiv:2310.06114](#)
- Yang, J., Glossop, C., Bhorkar, A., Shah, D., Vuong, Q., Finn, C., & Levine, S. (2024). Pushing the limits of cross-embodiment learning for manipulation and navigation. [arXiv:2402.19432](#)
- Yang, J., Cao, Deng, C., Antonova, R., Song, S., & Bohg, J. (2024). Equibot: Sim (3)-equivariant diffusion policy for generalizable and data efficient learning. [arXiv:2407.01479](#)
- Yoo, U., Hung, A., Francis, J., Oh, J., & Ichnowski, J. (2024). Ropotter: Toward robotic pottery and deformable object manipulation with structural priors. In *2024 IEEE-RAS 23rd international conference on humanoid robots (humanoids)* (pp. 843–850)
- Yu, W., Peng, J., Yang, H., Zhang, J., Duan, Y., Ji, J., Zhang, Y. (2024). Ldp: A local diffusion planner for efficient robot navigation and collision avoidance. In *2024 IEEE/RSJ international conference on intelligent robots and systems (IROS)* (pp. 5466–5472).
- Yu, T., Quillen, D., He, Z., Julian, R., Hausman, K., Finn, C., & Levine, S. (2020). Meta-world: A benchmark and evaluation for multi-task and meta reinforcement learning. In *Conference on robot learning* (pp. 1094–1100)
- Yu, T., Xiao, T., Stone, A., Tompson, J., Brohan, A., Wang, S., Singh, J., Tan, C., Peralta, J., Ichter, B., et al. (2023). Scaling robot learning with semantically imagined experience. [arXiv:2302.11550](#)
- Yuan, W., Dong, S., & Adelson, E. H. (2017). Gelsight: High-resolution robot tactile sensors for estimating geometry and force. *Sensors*, 17(12), 2762.
- Ze, Y., Zhang, G., Zhang, K., Hu, C., Wang, M., Xu, H. (2024). 3D diffusion policy. [arXiv:2403.03954](#)
- Zhang, F., & Gienger, M. (2024). Affordance-based robot manipulation with flow matching. [arXiv:2409.01083](#)
- Zhang, X., Chang, M., Kumar, P., & Gupta, S. (2024). Diffusion meets dagger: Supercharging eye-in-hand imitation learning. [arXiv:2402.17768](#)
- Zhang, L., Rao, A., & Agrawala, M. (2023). Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 3836–3847)
- Zhang, Z., Zhou, L., Liu, C., Liu, Z., Yuan, C., Guo, S., Tay, F.E. (2024). Dexgrasp-diffusion: Diffusion-based unified functional grasp synthesis method for multi-dexterous robotic hands. [arXiv:2407.09899](#)
- Zhang, Q., Liu, Z., Fan, H., Liu, G., Zeng, B., & Liu, S. (2024). Flowpolicy: Enabling fast and robust 3d flow-based policy via consistency flow matching for robot manipulation. [arXiv:2412.04987](#)
- Zhao, Q., Lu, Y., Kim, M.J., Fu, Z., Zhang, Z., Wu, Y., Li, Z., Ma, Q., Han, S., Finn, C., et al. (2025). Cot-vla: Visual chain-of-thought reasoning for vision-language-action models. In *Proceedings of the computer vision and pattern recognition conference* (pp. 1702–1713).
- Zhen, H., Qiu, X., Chen, P., Yang, J., Yan, X., Du, Y., & Gan, C. (2024). 3d-vla: A 3d vision-language-action generative world model. [arXiv:2403.09631](#)
- Zheng, Z., Peng, X., Yang, T., Shen, C., Li, S., Liu, H., & You, Y. (2024). Open-sora: Democratizing efficient video production for all. [arXiv:2412.20404](#)
- Zhou, S., Du, Y., Chen, J., Li, Y., Yeung, D. Y., & Gan, C. (2024). Robodreamer: Learning compositional world models for robot imagination. [arXiv:2404.12377](#)
- Zhu, F., Wu, H., Guo, S., Liu, Y., Cheang, C., & Kong, T. (2024). Irasim: Learning interactive real-robot action simulators. [arXiv:2406.14540](#)

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